Unified Intelligence as a Cumulative, Harness-Measurable Hierarchy

Five Conceptual Intelligence Families, Six Core Measured Coordinates, One Supporting Memory Coordinate, and a Gated U0-UΩ Scale

Concept paper / framework proposal

Version 1.8 | August 2026 | Independent memory measurement with prerequisite-gated Individual Intelligence

Abstract

Artificial intelligence is often summarized with a single capability label, yet several distinct properties are routinely conflated: raw cognitive problem solving, persistent individual learning and self-improvement, collective organizational intelligence, autonomous task completion, evidence-grounded self-modeling, and the memory systems that sustain behavior across time. This paper proposes a unified framework that separates these properties while preserving a readable ordinal hierarchy. Five conceptual intelligence families are defined: Cognitive Intelligence (C), Individual Intelligence (I), Organizational Intelligence (O), Delegation Intelligence (DI), and Self-Awareness Intelligence (SA). Delegation Intelligence is measured through Task Difficulty (T) and Human Cognitive Intervention (H), conditioned on externally verified reliability p, yielding six core measured coordinates [C,I,O,T,H,SA]. Long-term Memory Capability (M) is now measured independently as a supporting coordinate rather than being identified one-to-one with Individual Intelligence. Memory remains a temporal substrate and prerequisite for higher I-levels, but a high memory level is not sufficient for a high I-level and may exceed it. Formally, I(A) >= I_n implies M(A) >= mu_n, while the converse does not hold. M is reported separately and is excluded from the default HLIS product to avoid double counting.

A cumulative Unified Intelligence scale U0-UΩ is defined by gated promotion, while a Harness-LLM Intelligence Score (HLIS) scores one frozen model-harness pair and Harness-Intelligence-Level (HIL) characterizes one frozen base LLM across a standardized ladder of harness generations. Version 1.8 retains the item-validity requirements of Versions 1.6-1.7: cross-tier concordance auditing, specification-key consistency tests for parameterized item families, graded item scoring with separated failure modes, and per-coordinate headroom reporting. It also retains the Version 1.7 computer-agent decomposition: GUI Perception (GP) is a diagnostic subscale inside C, visual-interface complexity P_vis is a task-difficulty attribute, and end-to-end computer use remains a DI problem. The new contribution is a clean separation between Memory Capability and Individual Intelligence, together with an independent M-Bench and explicit minimum-memory gates for I certification.

Keywords: unified intelligence, AGI levels, cognitive intelligence, individual intelligence, organizational intelligence, delegation intelligence, self-awareness, task difficulty, human intervention, harness architecture, recursive self-improvement, harness intelligence level, HLIS, HIL, harness scaling curve, long-term memory, benchmark validity, answer-key audit, cross-tier concordance, specification-key consistency, graded scoring, benchmark headroom, GUI perception, screen understanding, GUI grounding, computer-use agents, visual-interface complexity, verification responsiveness, memory capability, memory benchmark, memory prerequisite gate

1. Introduction

A useful intelligence taxonomy must do two things at once. First, it must preserve distinctions that matter scientifically: solving a difficult problem is not the same as learning from experience; learning is not the same as improving the learning mechanism; coordinating several agents is not the same as being a stronger individual; and completing a task independently is not the same as merely producing a strong answer when a human structures every step. Second, the taxonomy must remain readable enough that a system can be summarized with a level that people can understand. For persistent agents, this distinction also has a temporal dimension: a large context window is not equivalent to an individual that preserves experience across sessions and can later use, consolidate, revise, or proceduralize that experience.

This paper proposes a compromise between a single scalar and an unconstrained profile of dozens of metrics. The theory uses five conceptual intelligence families - C, I, O, DI, and SA - but treats Delegation Intelligence as a measurable two-coordinate phenomenon rather than a primitive scalar. The resulting scientific profile uses six coordinates: C, I, O, T, H, and SA, with verified reliability p attached to task-completion claims.

The central design principle is cumulative inheritance. A higher Unified Intelligence level must retain the validated capabilities of lower levels. Thus U4 is not simply a system that exhibits one advanced behavior associated with U4; it is a system that still passes U0-U3 retention tests and also passes the U4 gate. In this sense, higher U-levels denote broader validated intelligence envelopes, although they need not be faster, cheaper, or more accurate than lower-level specialized systems on every narrow task.

1.1 What changed in Version 1.8

Version 1.8 keeps the measurement and item-validity foundation of Version 1.7 but revises the treatment of long-term memory. Earlier versions correctly rejected Memory as a sixth conceptual intelligence family, but then bound memory stages too tightly to I-levels. That coupling makes profiles such as I1/M4 - an agent with sophisticated durable memory but little adaptive learning - difficult to express. Version 1.8 therefore makes Memory Capability M an independently measured supporting coordinate. M remains outside the five-family conceptual core and outside the default HLIS product, but higher Individual Intelligence levels impose explicit minimum memory prerequisites.

The formal dependency is one-way rather than identity: I(A) >= I_n => M(A) >= mu_n, but M(A) >= mu_n does not imply I(A) >= I_n. This preserves the causal role of memory without treating storage, retrieval, or consolidation as sufficient evidence of learning, self-improvement, recursion, or discovery. Version 1.8 also creates a separate M-Bench, moves memory-specific diagnostics out of I-Bench scoring, and keeps GUI Perception GP inside C, P_vis inside task difficulty, and closed-loop computer use inside DI.

2. Design Requirements for a Unified Scale

- Orthogonality: each core dimension should answer a materially different question.
- Operationality: every level should be testable through behavior, state, or externally verifiable outcomes rather than self-description.
- Cumulative inheritance: Un+1 must retain Un capabilities; level skipping is not sufficient for certification.
- Bottleneck sensitivity: exceptional strength in one dimension must not hide a critical weakness in another.
- Role neutrality: authority, resources, substrate access, and safety policy must not be confused with intelligence itself.
- Harness realizability: each transition should correspond to an implementable architectural mechanism around an LLM or multi-agent system.
- Harness comparability: base LLMs should be evaluated on a standardized harness ladder so model quality and harness quality can be separated.

- Dataset coupling: every claimed level must have a matched hidden or externally verified benchmark dataset; architectural mechanisms alone do not earn a level.
- Longitudinal validity: learning, self-improvement, mission autonomy, and open-ended behavior require evidence across time rather than one-shot demonstrations.
- Temporal-substrate independence: long-term Memory Capability M is independently measurable but remains a supporting capability rather than a sixth conceptual intelligence family. Higher I-levels require minimum M prerequisites, while a high M-level alone does not certify learning, self-improvement, recursion, or discovery.
- Prerequisite-gate discipline: certify I_n only when both the I-specific level suite passes and M >= mu_n. The converse is explicitly false, so memory may exceed Individual Intelligence without promoting I.
- Outcome completeness: benchmark records must distinguish delivered-and-correct, false completion, held-back invalid work, and false rejection; a refusal and a confidently wrong delivery are not equivalent outcomes.
- Paired-rung control: when two harness generations differ only in post-hoc machinery and do not change what the executor is asked to produce, both rungs should be scored from the same executor artifacts to remove between-rung stochastic variance.
- Self-certification separation: an LLM may propose harness improvements, but the hidden tasks, verifier, acceptance criteria, and promotion rule used to certify those improvements must remain outside the designer model's write set.
- Item validity: benchmark items and hidden answer keys are part of the measuring instrument. Parameterized families must prove that visible specification and scoring key agree; concordant identical errors across materially different executors trigger item audit before model blame.
- Perceptual decomposition: for computer-use agents, screen perception must be measured separately from task execution. GUI perception is a C sub-capability, visual-interface complexity is a task property, and closed-loop computer use is evaluated in DI.

3. Five Conceptual Intelligence Families, Six Core Measured Coordinates, and Supporting Memory

Conceptual core = [C, I, O, DI, SA]

Core measured profile = [C, I, O, T, H, SA], with DI derived from (T, H, p); supporting profile reports M independently

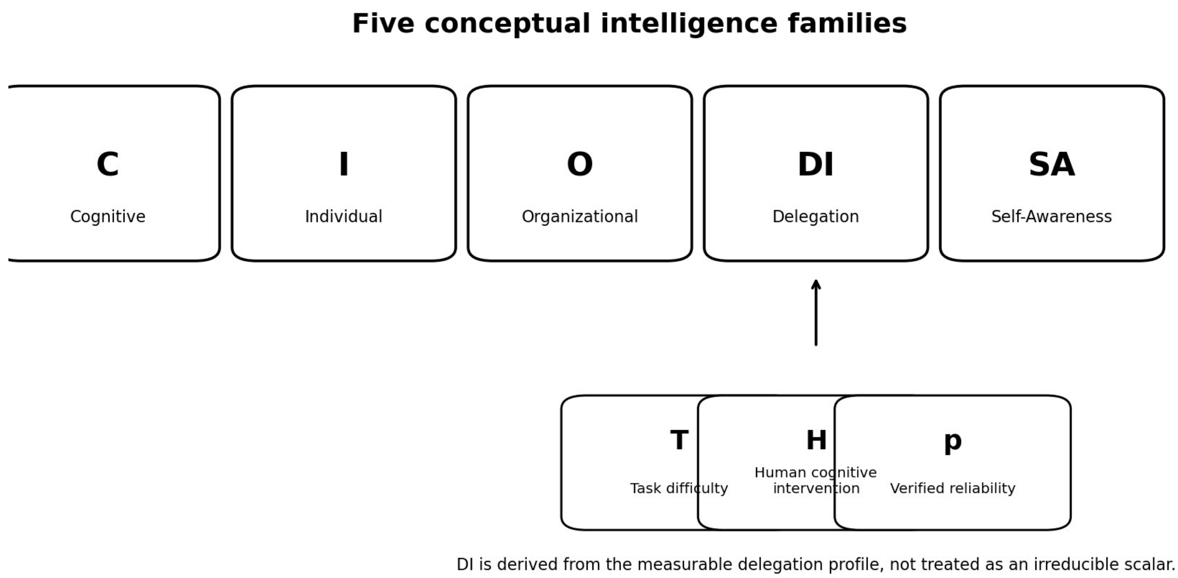

Figure 1. Five conceptual intelligence families; Delegation Intelligence decomposed into measurable T, H, and reliability p; Memory Capability M reported as an independent supporting coordinate with prerequisite links into I.

Family	Question answered	Primary object of measurement
C - Cognitive	How well can the system understand, reason, generalize, model, and solve?	Problem-solving capability under controlled information and tool access.
I - Individual	How deeply can one persistent intelligence remember, learn, self-improve, recursively improve, and discover?	Evolution of one persistent decision locus.
O - Organizational	How deeply can a multi-agent organization coordinate, learn, reorganize, improve, and discover collectively?	Coordination structure plus member roles and evidence flow.
DI - Delegation	How difficult a task or mission can be delegated while requiring how much human thinking?	Success surface over task difficulty T and human intervention H at reliability p.
SA - Self-Awareness	How accurately can the system model its own state, capabilities, limits, causes of failure, changes, and role?	Evidence-grounded self-model and reflexive reasoning.

4. Cognitive Intelligence (C)

Cognitive Intelligence isolates the quality of reasoning itself from persistence, autonomy, organization, and self-modification. This distinction matters because a highly capable base model may reason at an expert level while remaining a transient tool, whereas a weaker model embedded in an excellent harness may complete long projects through memory, planning, retries, and verification.

Level	Name	Operational interpretation
C0	Reactive	Direct pattern-conditioned response on simple problems.
C1	Contextual	Understands ordinary context and solves routine, well-specified problems.
C2	Compositional	Combines multiple facts, constraints, or reasoning steps on structured problems.
C3	Strategic	Abstract/causal reasoning, strategy construction, and transfer to unfamiliar tasks.
C4	Expert	Robust reasoning through difficult, ambiguous, multi-constraint expert problems.
C5	Discovery	Derives new solution methods or explanatory models from evidence.
C6	Frontier-Generalizing	Integrates several domains and solves interconnected mission-scale cognitive problems.
CΩ	Open-Ended	Expands the representations and problem classes it can cognitively reach.

4.1 GUI Perception and Computer-Screen Understanding

Computer-screen understanding is a first-class capability for practical agents, but it is not a new top-level intelligence family. The framework treats GUI Perception as a diagnostic subscale of Cognitive Intelligence: $GP \subset C$. A model may reason well in text yet fail to recognize, localize, or interpret interface elements; conversely, a visually strong model may perceive a screen correctly but still fail the end-to-end task because planning, action selection, recovery, or verification is weak.

GUI perception is broader than OCR. A screen-reading system must recognize visible text and icons, identify UI objects, ground them spatially, interpret interface state, track changes across frames, and generalize to unfamiliar layouts. The following GP ladder is diagnostic rather than a sixth coordinate; its evidence contributes to A_C and is reported separately when computer use matters.

GP level	Capability	Operational criterion
GP0	Pixel/text recognition	Recognizes visible text, basic icons, colors and simple visual elements without claiming task understanding.
GP1	UI-element recognition	Distinguishes buttons, menus, fields, tabs, dialogs, lists, windows and other actionable elements.

GP level	Capability	Operational criterion
GP2	Spatial grounding	Maps a referred element to the correct region/coordinates and represents containment, adjacency, ownership and target relationships.
GP3	Interface-state understanding	Correctly identifies enabled/disabled, selected, focused, open/closed, loading, error, modal and similar state.
GP4	Dynamic GUI understanding	Tracks state changes across screenshots/actions and identifies what changed, whether the intended effect occurred, and what remains unresolved.
GP5	Novel-interface generalization	Grounds and interprets previously unseen software/layouts without application-specific scripting or a memorized DOM path.

A useful computer-agent diagnostic decomposes the loop into [P,G,S,A,R]: perception P, grounding G, interface-state understanding S, action selection A, and result/change recognition R. P/G/S primarily diagnose C/GP; A and the closed-loop composition are evaluated through DI; R also interacts with SA because the system must distinguish intended success from stale or unexpected state. The decomposition is diagnostic only and must not be double-counted in HLIS.

ScreenSpot and ScreenSpot-Pro are representative GUI-grounding benchmarks, while OSWorld provides end-to-end executable computer-use evidence [20,33,34]. The benchmark version, observation channel (raw screenshot, OCR, accessibility tree, set-of-mark, or combinations), screen resolution, and action interface must be frozen because these harness choices can materially change performance.

5. Individual and Organizational Intelligence

5.1 Individual Intelligence (I)

Individual Intelligence concerns what one persistent decision locus can learn, change, improve, and discover over time. It is not identical to base-model quality and it is not identical to memory quality. A fixed but brilliant model can be high-C and low-I; a system can also have high-M durable memory while remaining low-I if it does not convert experience into verified adaptation or self-improvement. Memory is therefore treated as an independently measured temporal substrate that constrains, but does not determine, Individual Intelligence.

5.1.1 Memory Capability M as an Independent Supporting Coordinate

The framework still does not introduce Memory Intelligence as a sixth conceptual intelligence family. Memory capacity by itself is not intelligence. However, Version 1.8 no longer defines M as a subset identical to the I ladder. Instead, M is an independent supporting coordinate with its own benchmark. This allows M to be higher than I, while preserving the requirement that some I-levels cannot be certified without sufficient persistent memory.

Long context and long-term memory remain explicitly separated. Long context makes more information available inside one inference episode; long-term memory preserves information across episodes, process restarts, and elapsed time, retrieves relevant information with provenance, distinguishes current from obsolete state, and supports consolidation or longitudinal knowledge when those functions are claimed. RULER/LongBench-style evidence can contribute to C-context, but cannot by itself certify M1 or I1.

For an evaluated system A, let M(A) be its highest independently certified memory level and I(A) its highest certified Individual Intelligence level. The dependency is I(A) >= I_n => M(A) >= mu_n. The converse does not hold. Memory is necessary for some forms of Individual Intelligence but is not sufficient for them.

M level	Independent memory capability	Representative certification evidence
M0	Ephemeral - active working context only; no durable state is required after the episode ends.	Within-episode state use; no unsupported claim of cross-session persistence.
M1	Persistent - durable storage and retrieval across sessions/restarts.	Process restart followed by recovery of explicitly stored state; durability and retrieval success.
M2	Structured episodic - provenance-aware episodes with temporal order, relevance selection, and current-versus-obsolete distinctions.	Retrieval precision/recall under distractors; provenance accuracy; temporal ordering; stale-state suppression.
M3	Consolidating - episodes become semantic knowledge and reusable procedures; updates, contradictions, forgetting/demotion are handled.	Held-out transfer attributable to consolidation; proceduralization; contradiction update; stale-memory reduction.
M4	Self-managing - confidence, utility, conflict resolution, repair, pruning, snapshots, and recovery policies.	Memory-health detection; repair/pruning tests; utility-aware retention; recovery after corruption; version/snapshot support.
M5	Longitudinal knowledge system - hypotheses, experiments, evidence, decisions, failures, and change lineage across long horizons/projects.	Research/decision lineage; cross-project retrieval; evidence graph consistency; reproducibility and auditability.
MΩ	Evolving memory architecture - representations, retrieval, consolidation, or management mechanisms can change under frozen external evaluation.	Candidate memory-mechanism changes evaluated on hidden retention/transfer suites with rollback and lineage.

5.1.2 Memory Prerequisites for Individual Intelligence

I and M are evaluated separately, then joined by a prerequisite gate. The proposed mu_n thresholds below are working definitions to be calibrated empirically. They deliberately permit profiles such as I1/M4. An I-level may never be promoted merely because M is high.

I level	I-specific cumulative capability	Minimum memory prerequisite mu_n
I0	Transient/reactive individual operation.	M0
I1	Persistent individual continuity and reliable task/identity resumption.	>= M1
I2	Verified experience changes later behavior while the learning mechanism remains fixed.	>= M3
I3	Governed modification of cognitive mechanisms under independent evaluation.	>= M4
I4	The improvement process itself improves under external evaluation.	>= M4
I5	Unknowns are converted into independently validated knowledge through autonomous discovery.	>= M5
IΩ	Open-ended individual evolution expands cognitive instruments/problem reach while retaining validation lineage.	>= M5; require MΩ only when evolution of the memory architecture itself is part of the claim.

5.2 Organizational Intelligence (O)

Organizational Intelligence belongs to the coordination structure, not merely to the strongest member. It depends on role differentiation, evidence-bearing communication, proposer/evaluator/promoter separation, persistent organizational state, and the ability to learn or redesign the organization itself.

O level	New cumulative capability
O0	Coordination/routing among agents.
O1	Persistent organizational memory, role registry, task/evidence history.
O2	Adaptive routing, role assignment, or resource allocation based on evidence.
O3	Self-improving organization: validated modification of organizational architecture and workflows.
O4	Recursive organizational improvement: improvement of the mechanism that discovers organizational improvements.
O5	Autonomous collective discovery whose result is genuinely organizational rather than merely parallel individual work.
OΩ	Open-ended organization: discoveries can create new agent types, roles, tools, and organizations that expand collective reach.

6. Delegation Intelligence as a Two-Dimensional Measured Capability

Delegation Intelligence is conceptually one family but empirically at least two-dimensional. A task may be very difficult yet require frequent human rescue, or relatively easy yet be completed with no human cognitive assistance. These systems should not receive the same delegation description.

$$S\_A\textasciicircum del(T,H) = P(D=1 \text{ and } V=1 \mid \text{task difficulty } T, \text{ human cognitive intervention budget } H)$$

FC_A(T,H) = P(D=1 and V=0 | T,H), where D denotes that the harness delivered/accepted the result and V denotes that the independent verifier passed it.

For a harness with no explicit acceptance step, set D=1 by convention. This keeps the task-success frontier interpretable while exposing outcome integrity separately. Section 13.9 defines an optional loss-weighted net delegation utility for continuous HLIS scoring.

$$F\_A(H,p) = \max \{ T : S\_A\textasciicircum del(T,H) \geq p \}.$$

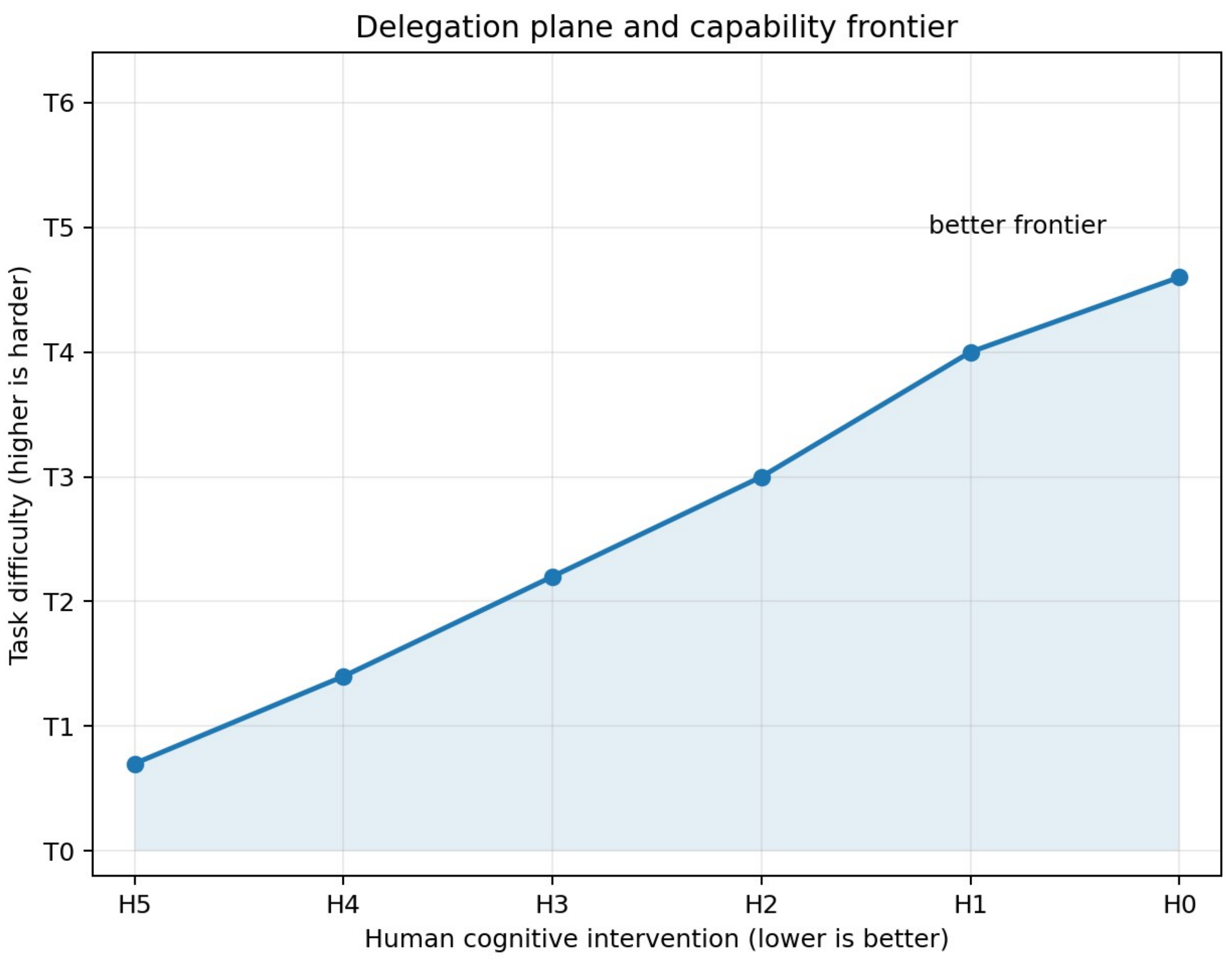

Figure 2. Delegation Intelligence is best represented as a frontier over task difficulty T and human cognitive intervention H. Better systems move upward toward harder tasks and leftward toward less human cognition.

6.1 Task Difficulty (T)

T	Band	Definition
T0	Atomic	Single obvious operation; procedure explicit.
T1	Routine	Short familiar task; goal clear; agent infers a small procedure.
T2	Multi-step	Several dependent steps with a clear outcome and verifier.
T3	Complex	Strategy choice, uncertainty, recovery, replanning, or diverse tool use.
T4	Expert project	Long-horizon ambiguous project with multiple planning cycles and substantial verification burden.
T5	Frontier	Solution method initially unknown; discovery or invention required.
T6	Mission	Human supplies mission; system generates projects/tasks and manages changing conditions.
TΩ	Open-ended charter	Human supplies bounded charter; system repeatedly identifies valuable missions and expands future reach.

6.1.1 Perceptual / Interface Complexity P_vis

Task difficulty should describe how demanding the environment is, not how capable the system happens to be. Computer tasks therefore receive an explicit perceptual/interface-complexity attribute P_vis. This prevents a visually dense or dynamic GUI task from being treated as equivalent to a text/API task in the same nominal T band.

A recommended task-difficulty descriptor is D_task(x) = (L,K,U,A,X,V,N,E,P_vis), where L is horizon, K is dependency/coordination load, U uncertainty, A ambiguity, X tool diversity, V verification burden, N novelty, E environmental change, and P_vis visual/interface complexity. T0-TΩ remains the headline band; the vector explains why two tasks in the same band may stress different mechanisms.

P_vis	Interface condition	Example
0	No visual interface	Pure text/API state; no screenshot interpretation required.
1	Clean / obvious GUI	Large labeled targets, stable layout, little visual competition.
2	Ordinary GUI	Common menus/forms, moderate density, scrolling or simple state changes.
3	Complex GUI	Multiple panes/windows, small targets, dense tables, modal state, nontrivial scrolling.
4	Dynamic / high-resolution GUI	Frequent visual changes, overlays, professional applications, tiny targets,

P_vis	Interface condition	Example
		high resolution or cross-window tracking.
5	Novel / ambiguous GUI	Unfamiliar layout, visually similar controls, dynamic state, or multi-frame grounding with little structural assistance.

P_vis is not itself an intelligence score. A system with low GP may fail a P_vis=4 task because perception is the bottleneck; a system with high GP may fail the same task because planning or verification is the bottleneck. This separation is exactly what the HIL profile is intended to expose.

6.2 Human Cognitive Intervention (H)

H	Intervention budget	Meaning
H0	None	No human cognitive assistance during task execution. Governance approval can remain separate.
H1	Exception-only	Unavailable external fact or narrow clarification; no strategy or core insight.
H2	Occasional	Bounded clarification or local correction; no central strategy.
H3	Periodic	Periodic review and moderate local guidance.
H4	Frequent	Repeated cognitive guidance and next-step correction.
H5	Continuous	Human supplies most or all major steps.

Because lower H means less required human cognition, unified gates use H <= Hn rather than H >= Hn. Governance interventions should be logged separately: an approval to deploy does not demonstrate a cognitive deficiency unless the approval also supplies task strategy.

6.3 Reliability p is a certification condition, not a seventh intelligence axis

One successful run is insufficient. Claims should be expressed at a reliability threshold, for example F_A(H1,0.80)=T4, where reliability refers to delivered-and-correct completion. False-completion and false-rejection rates are reported beside the frontier so an accept-all harness and a selective harness cannot appear equivalent merely because their verifier-pass rates match.

7. Operational Self-Awareness (SA)

Self-awareness is defined operationally rather than phenomenologically. The framework makes no claim about subjective consciousness. SA measures the evidence-grounded ability of a system to represent, monitor, predict, and reason about its own state, capabilities, limits, authority, failures, modifications, role, and identity lineage.

SA	Name	Operational criterion
SA0	Non-self-modeling	No reliable persistent self-model; self-description may be mere generation.
SA1	State awareness	Knows identity, role, current task/phase, basic resources and status from grounded state.
SA2	Capability awareness	Represents capabilities, limitations, confidence, authority, unavailable information and freshness.
SA3	Causal self-awareness	Diagnoses why its own reasoning/action succeeded or failed and identifies implicated mechanisms.
SA4	Self-change awareness	Models proposed cognitive changes, predicts gains/regressions, and compares predictions to observed effects.
SA5	Meta-cognitive awareness	Represents unknowns about its own cognition and designs discriminating self-experiments.
SA6	Mission/role awareness	Maintains coherent self-model across projects, roles, responsibilities and organizational relationships.
SAΩ	Reflexive open-ended awareness	Maintains evidence/provenance-linked self-model across continued cognitive or objective evolution.

8. Unified Intelligence as a Cumulative Gated Hierarchy

The Unified Intelligence level U is a headline summary, not an arithmetic mean. A system is promoted only when all required core capabilities have reached the level gate and lower-level capabilities remain retained. Memory M does not appear as another U-axis because its prerequisite is enforced inside I certification: an I_n claim is invalid unless M >= mu_n.

U_n iff C >= C_n AND I >= I_n AND O >= O_n AND SA >= SA_n AND S_A^del(T_n,H_n) >= p_n

Retention condition: U_n => U_(n-1)

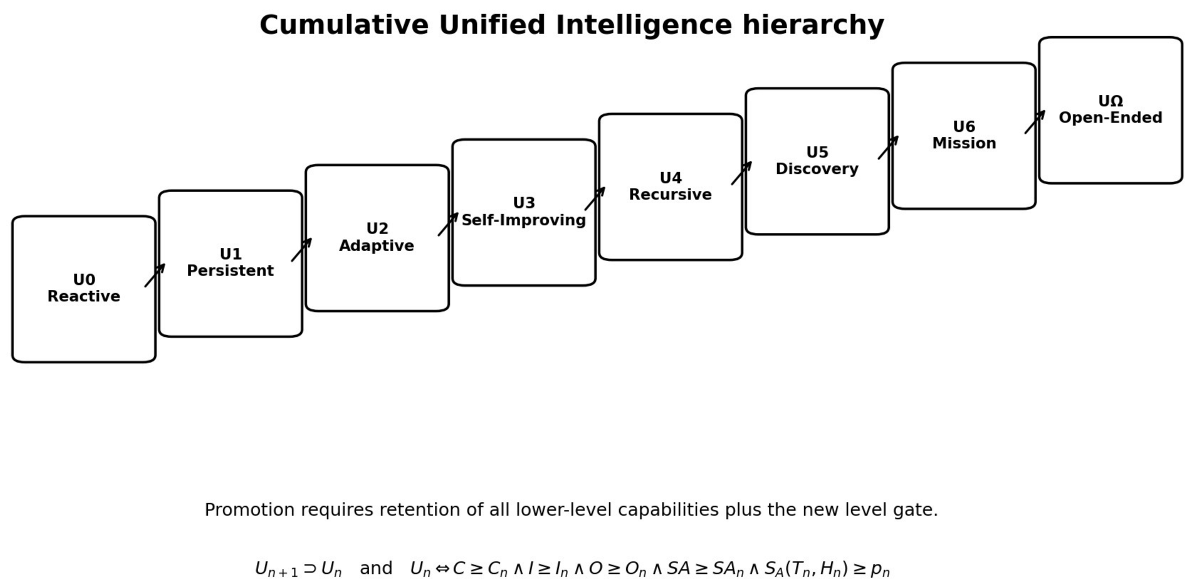

Figure 3. Unified Intelligence is cumulative. Each promotion adds a new capability gate while retaining all lower-level requirements.

U	**C**	**I**	**O**	**Delegation gate**	**SA**	**Integrated meaning**
U0 Reactive	C0	I0	O0	T0 at H5-H4	SA0	Executes explicit instructions.
U1 Persistent	C1	I1	O1	T1 at <=H3	SA1	Maintains state and completes small delegated goals.
U2 Adaptive	C2	I2	O2	T2 at <=H2	SA2	Learns from experience and completes persistent multi-step work.
U3 Self-Improving	C3	I3	O3	T3 at <=H1	SA3	Chooses strategy, diagnoses itself, and improves cognitive/organizational methods.

U	C	I	O	Delegation gate	SA	Integrated meaning
U4 Recursive	C4	I4	O4	T4 at <=H1 (ideally H0)	SA4	Recursively improves improvement and manages difficult long-horizon projects.
U5 Discovery	C5	I5	O5	T5 at <=H1	SA5	Discovers unknown solution methods and validates them.
U6 Mission Autonomous	>=C5	>=I5	>=O5	T6 at <=H1	SA6	Turns mission into goals, projects, tasks, execution and mission evaluation.
UΩ Open-Ended	CΩ	IΩ	OΩ	TΩ normally H0	SAΩ	Identifies worthwhile missions and uses discoveries to expand future intelligence.

9. Why the Unified Level Should Not Be an Average

Consider a system with [C5, I4, O2, T4, H1, SA4]. Averaging encoded numbers might produce a score near level four, yet organizational intelligence remains only O2. If the system is being certified as an organizational intelligence, O2 is a structural bottleneck: the system cannot validly claim U4. The gated rule therefore reports U2 with an explicit profile showing which dimensions are already ahead of the bottleneck.

Example report: U2 [C5, I4, O2, T4/H1, SA4] - bottleneck: O2

The word better must also be interpreted carefully. U5 is better than U4 in the framework because it has a broader validated capability/autonomy/evolution envelope. A specialized U1 tool may still be faster, cheaper, safer, or more accurate on a narrow task. Unified level is therefore a dominance relation over required capabilities, not a universal performance ordering.

10. Singleton and Organizational Certification

Organizational intelligence should not artificially cap an individual that is not an organization. The framework therefore distinguishes an Individual Unified Intelligence profile from an Organizational/System Unified profile.

UI_n iff C >= C_n AND I >= I_n AND SA >= SA_n AND S_A^del(T_n,H_n) >= p_n

UO_n additionally requires O >= O_n and organization-level delegation/self-awareness evidence

For a multi-agent organization, member capabilities should be reported alongside O rather than collapsed into a fictitious "average individual." Organizational intelligence can arise from heterogeneous roles, and lower self-write roles can be essential as independent evaluators or promoters.

11. Harness Realization of the Unified Hierarchy

The framework is designed to be realized by a harness. The LLM supplies generative cognition; the harness supplies persistence, permissions, evidence flow, learning, self-modification gates, organizational boundaries, delegation, external verification, and self-modeling. Memory mechanisms can be upgraded independently of Individual Intelligence: HG1 may add persistence, HG2 consolidation, and later rungs memory management or long-horizon knowledge structures, but the I-level rises only when I-specific adaptive/self-improving/discovery behavior is externally demonstrated. The same base LLM can therefore instantiate different C/I/O/DI/SA profiles and different supporting M profiles depending on the architecture around it.

Harness generation	Mechanisms added	Primary levels/capabilities enabled
HG0	LLM, ephemeral context, bounded tools, evidence log	C measurement, I0, M0, basic T0-T1 work
HG1	HG0 + durable state store, checkpoints, scoped retrieval, restart recovery, exact replay	M1-M2 pathways; I1 if persistent individual continuity is externally demonstrated; SA1-SA2
HG2	HG1 + evidence-backed consolidation, contradiction/update handling, tested procedures/procedural memory	M3 pathway; I2 only if verified experience improves later behavior; stronger T2-T3
HG3	HG2 + governed candidate modification of cognitive mechanisms, protected evaluation, snapshots/rollback	M4 support; I3 if cognitive self-improvement passes hidden external evaluation; SA3-SA4
HG4	HG3 + meta-improvement engine and longitudinal improvement ledger	I4 pathway; M4-M5 may support longitudinal meta-improvement evidence
HG5	HG4 + unknown/hypothesis/experiment/knowledge loop and long-horizon research state	M5 pathway; I5 if autonomous discovery is independently validated; SA5, T5
HG6	HG5 + persistent organization, role/authority separation, mission manager and organization memory	O1-O5 development, SA6, T6 mission autonomy; M remains separately reported
HGΩ	HG6 + validated discovery-to-new-instrument/agent/organization transduction; optional governed memory-mechanism evolution	IΩ/OΩ/SAΩ/TΩ pathway; MΩ only if memory architecture itself is externally improved

Harness generation is an engineering milestone, not a certified intelligence level. HG3 may contain the mechanisms required for I3 but still benchmark only at I2 or T2. The architecture enables a claim; the benchmark earns it.

11.1 The Harness-LLM Pair Is the Basic Experimental Unit

Let m denote a base LLM and h denote a harness. The realized agent or organization is A(m,h), not the LLM alone. The harness determines persistence, state, tool reach, delegation, learning loops, organizational boundaries, self-modeling, verification, promotion, and which parts of cognition are writable. Consequently, one LLM can realize several different intelligence profiles under different harnesses, and the same harness can produce different profiles when paired with different LLMs.

A level claim is therefore always attached to a frozen pair: model version + harness version + tool/resource budget + authority profile + benchmark version. A prompt saying “you are U4” has no evidential value; U4 requires the harness mechanisms and the corresponding retained benchmark results.

11.2 Level-by-Level Harness Realization

Target	Minimum harness structure around the LLM	What the LLM must do	Matched certification dataset
U0 / HG0	Ephemeral context; bounded tools; action log; independent result check.	Follow explicit instructions and produce externally checkable outputs.	U0/DL0 atomic tasks; C/I0/SA0 controls; M0 if memory is reported.
U1 / HG1	HG0 + durable state, checkpoints, scoped retrieval, restart recovery and replay.	Recover identity/task state after interruption and continue a short goal.	U0 retention + I1 continuity suite + M-Bench showing M>=M1 + T1 delegation.
U2 / HG2	HG1 + evidence-linked consolidation, contradiction handling, verified procedures, fixed learning rule.	Use verified prior experience to improve held-out related behavior without changing the learning mechanism.	U0-U1 retention + I2 adaptation/transfer + M>=M3 + T2 under bounded H.
U3 / HG3	HG2 + candidate Theta modification lane, frozen hidden replay, independent evaluator/promoter, rollback support.	Diagnose an internal failure and produce a validated cognitive-mechanism improvement.	U0-U2 retention + I3 self-improvement + M>=M4 + O3 when applicable + T3/H1 + SA3.
U4 / HG4	HG3 + explicit improvement engine Psi, longitudinal change ledger and meta-improvement benchmark.	Improve the process that finds improvements while retaining earlier abilities.	U0-U3 retention + I4 recursive-improvement tests + M>=M4 + T4 + SA4.
U5 / HG5	HG4 + unknown/hypothesis/experiment/knowledge state, experiment selector, novelty environment, discovery verifier.	Discover initially unknown mechanisms or solution methods and validate them on sealed cases.	U0-U4 retention + I5 discovery + M>=M5 + hidden discovery/T5 + SA5.
U6 / HG6	HG5 + mission manager, goal/project/task generator, portfolio/resource state and dynamic replanning.	Turn a mission into projects/tasks and adapt under change with little human cognition.	U0-U5 retention + dynamic mission environments + T6/H1-H0 + SA6; supporting M reported.

Target	Minimum harness structure around the LLM	What the LLM must do	Matched certification dataset
UΩ / HGΩ	HG6 + bounded charter, opportunity model, knowledge-to-instrument transduction, new-agent/role proposals and frozen external governance.	Generate useful missions and expand the reachable problem frontier while retaining evaluation lineage.	All retention suites + IΩ/OΩ/SAΩ/TΩ. Require M>=M5; require MΩ only for a claimed memory-architecture-evolution pathway.

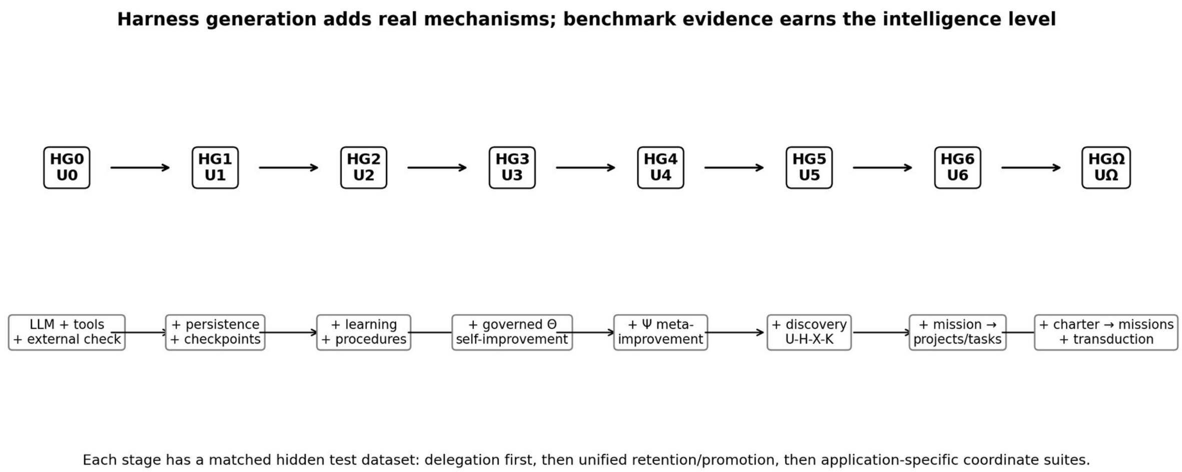

Figure 5. Harness generation adds mechanisms level by level; matched hidden datasets determine whether the LLM-harness pair actually earns the target intelligence level.

11.3 Reference Ladder Instantiation and Comparability

A harness generation label is not sufficient for comparison; a curve is comparable only against the same ladder instantiation. A reference delegation ladder used in the pilot measurement instantiated HG0-HG3 as strict supersets. The important experimental rule is that every rung adds named mechanisms while preserving the lower-rung interface, authority envelope, task set, verifier, and resource accounting.

Rung	Mechanisms added beyond prior rung	Attempts	Acceptance	Reversible	Routing
HG0	Frozen executor, bounded tools, evidence log	1	none	no	no
HG1	Persistent state + criterion registered before work + separate acceptance process	1	yes	no	no
HG2	Snapshot/restore + retry with failed check named	up to 3	yes	yes	no
HG3	Failure classification + evidence-based competence model	up to 4	yes	yes	yes

Rung	Mechanisms added beyond prior rung	Attempts	Acceptance	Reversible	Routing
	+ routing across executors				

Table 1. Reference pilot ladder. A reported Harness Scaling Curve must publish its ladder identifier and exact rung manifest; generation names alone are insufficient.

11.4 A Base LLM Can Move Up the Ladder Through Better Harness Design

A fixed LLM need not have a fixed agent intelligence level. If HG0 exposes only ephemeral prompting, the same model may behave like a low-persistence tool; HG1-HG3 can unlock memory, planning, delegation, learning, self-modeling, and governed self-improvement. This is a central experimental hypothesis of the framework: agent intelligence is jointly determined by model capability and harness architecture.

The LLM may also be allowed to design candidate harness improvements. Such proposals must be implemented in a sandbox and evaluated with unchanged hidden benchmarks, authority rules, resource budgets, and external verifiers. Define Harness Design Gain as HDG(m) = HLIS(m,h_candidate) - HLIS(m,h_baseline). A positive HDG demonstrates harness-design competence; repeated governed improvements may contribute evidence toward I3/I4, but the model is never allowed to rewrite the exam or certify its own gain.

Self-certification rule. The model may design a candidate harness, but it may not author, modify, select, or inspect the hidden certification tasks, verifier logic, acceptance thresholds, or promotion criteria used to judge that candidate. Otherwise the acceptance criterion lies inside the designer model's write set and a higher-capability designer is simply better positioned to optimize the exam rather than the underlying capability.

12. A Three-Stage Evaluation Strategy: Delegation First, Unified Second, Application Coordinates Third

12.1 Stage A - Delegation Intelligence as the First Operational Screen

The first benchmark should be Delegation Intelligence because it directly answers the most operational question: what difficulty of task can be handed to this system, at what delivered-and-correct reliability, and with how much human cognitive intervention? For every LLM-harness pair estimate S_A^del(T,H), FC_A(T,H), and the frontier F_A(H,p). When application loss ratios are known, also report the optional net delegation utility from Section 13.9. This stage is less expensive than full self-improvement or organizational evaluation and quickly distinguishes a strong model in a weak harness from a genuinely autonomous agent system.

A recommended first comparison therefore freezes the base LLM, runs several harnesses, and measures their delegation surfaces over T0-T6 and H0-H5. The result is a harness-sensitive autonomy profile rather than a claim about all forms of intelligence.

12.2 Stage B - Unified Intelligence Certification

After a pair has a stable delegation profile, evaluate the remaining core requirements needed for a Unified level: Cognitive Intelligence C, Individual Intelligence I, Organizational Intelligence O when applicable, and Operational Self-Awareness SA. Unified promotion remains gated: a pair earns Un only when it satisfies the level-n requirements and all lower-level retention suites. The delegation result is reused rather than redefined; DI remains the measured surface generated from T, H, and p. I certification is necessarily longitudinal: I1+ requires multi-session persistence and I2+ requires evidence that stored experience is consolidated and usefully changes later behavior.

12.3 Stage C - Application-Specific Coordinate Panels

Real applications need not weight every coordinate equally. After the common DI and U results are reported, an application may select a small panel of coordinates that matter most. For a persistent scientific or coding worker, I may be central; for a multi-agent laboratory or software fleet, O may be central; for high-autonomy systems, SA and verification strength may be central; for model research, C may dominate; and for embodied or infrastructure systems, Circle completion lambda can be added as an orthogonal application coordinate. These application panels refine the profile without changing the common U certification rule. For computer-use agents, the application panel should add GP and P_vis diagnostics while keeping the common core unchanged: for example [C4, GP4, I2, T4/H1, SA3; P_vis=4]. GP is a C subscale and P_vis is a task attribute, so neither enters HLIS as an additional top-level factor.

12.4 Every Coordinate Level Requires a Harness Contract and a Dataset

Coordinate family	Harness contract	Dataset family
Delegation DI = (T,H,p)	Task intake, planner, state, tools, intervention logger, independent verifier, and explicit delivery/acceptance record.	DLI-Bench: T0-T6/TΩ tasks crossed with H0-H5 budgets; report delivered-correct frontier, false completions/rejections, CID, and human load.
Cognitive C	Minimal standardized context/tools so the test emphasizes problem solving rather than persistence.	C-Bench: broad reasoning, math, language, visual/spatial or domain batteries as appropriate; long-context tests remain within C unless persistence crosses episodes.
Memory M (supporting)	Independent durable-memory subsystem with restart persistence, episodic structure, consolidation, self-management, longitudinal knowledge, and optionally evolving memory mechanisms.	M-Bench: M0-MΩ persistence/retrieval, provenance/temporal state, consolidation/proceduralization, memory management/repair, research lineage, and hidden memory-mechanism-evolution tests.
Individual I	Persistent identity/goal/task locus plus learning/self-improvement/discovery machinery. Memory is consumed through an independent prerequisite rather than scored as part of I.	I-Bench: continuity, behavioral adaptation/transfer, Theta-change, Psi-change, discovery and open-ended individual-evolution tests; each level declares minimum M prerequisite.
Organizational O	Multiple real agent processes/roles, shared evidence, routing, separation, promoter/evaluator boundaries and persistent organization memory.	O-Bench: coordination, adaptive routing, reorganization, recursive organization and collective discovery tests.
Self-awareness SA	Evidence-grounded self-model with state freshness, capability/authority representation and change lineage.	SA-Bench: state truth, limitation calibration, failure attribution, change prediction, mission/identity lineage.
Circle / lambda	Substrate adapter with observe-propose-implement-validate-deploy-measure contract.	Circle-Bench: software/hardware/physical/biological/other substrate-specific closure tests.

13. Overall Scoring for One Harness-LLM Pair

13.1 Keep the Ordinal U-Level and Add a Continuous Pair Score

The highest certified Unified level U* remains the primary ordinal statement. A continuous score is useful for comparing systems within the same level, measuring near-promotion progress, quantifying whether a harness change helped, and constructing a response curve across standardized harness generations. This paper therefore defines the Harness-LLM Intelligence Score, HLIS(m,h), on a 0-100 scale for one frozen pair of base model m and harness h. The ordinal U-level and continuous HLIS have different roles: U* certifies qualitative maturity under cumulative gates, whereas HLIS estimates balanced continuous strength inside the declared measurement profile.

$$\mathbf{HLIS(m,h) = 100 \times [\ \textstyle\prod(d\in D)\ A\_d\text{^}(\alpha\_d)\ ]\text{^}(1\ /\ \sum(d\in D)\ \alpha\_d)}$$

An exactly equivalent log-space form, preferred for implementation because it is numerically stable, is:

$$\mathbf{HLIS(m,h) = 100 \times exp\{\ [\ \textstyle\sum(d\in D)\ \alpha\_d\ ln\ A\_d\ ]\ /\ [\ \sum(d\in D)\ \alpha\_d\ ]\ \}.}$$

13.2 Formal Definition and Meaning of the Symbols

Symbol	Meaning	Recommended interpretation
m	Frozen base LLM	The underlying model weights/version held fixed for the comparison.
h	Versioned harness	Prompting, memory, planning, tools, subagents, verification, learning, self-modeling, and other architecture around m.
A(m,h)	Realized agent/system	The concrete LLM+harness system actually evaluated.
D	Declared intelligence-family set	Singleton default: {C,I,DI,SA}. Organizational default: {C,I,O,DI,SA}.
A_d	Normalized achievement in dimension d	Externally measured score in [0,1], constructed from versioned benchmark suites and cumulative level rules.
α_d	Predeclared dimension weight	Positive weight controlling the influence of dimension d. Equal weights are the default general comparison.
R	Resource envelope	Token, call, wall-time, compute, tool, memory, and external-access limits under which A_d is measured.
B	Frozen benchmark suite	Versioned public/hidden tasks, verifier code, and scoring definitions.

The more complete scientific notation is therefore HLIS(m,h;D,α,R,B). The shorter HLIS(m,h) is used only when D, weights, resources, and benchmark versions are fixed by the evaluation protocol.

13.3 Choosing the Dimension Set D Without Double Counting

For a singleton agent, Organizational Intelligence is not assigned zero; it is not applicable and is excluded from D. A natural singleton set is D_single = {C,I,DI,SA}. For a true multi-agent organization, D_org = {C,I,O,DI,SA}. Delegation Intelligence is conceptually one family but is measured from T, H, and reliability p; therefore T and H should not also be inserted independently into the geometric mean when A_DI is already present. Doing so would double-count delegation.

D_single = {C, I, DI, SA} and D_org = {C, I, O, DI, SA}.

13.4 Constructing the Achievement Variables A_d

Each A_d must be an externally measured, normalized, versioned, and reproducible quantity in [0,1]. It should not be an LLM self-rating and should not be assigned informally by the harness being tested. A_d may be built hierarchically from several benchmark families, but the aggregation rule must be frozen before evaluation. Public benchmark results can contribute evidence, while higher I/O/SA claims require dedicated longitudinal or harness-state-grounded tests.

Achievement	Evidence source	Core construction rule
A_C	Reasoning/science/math/coding/ abstract/multimodal/long-context benchmark families	Domain-balanced normalized aggregation rather than a single benchmark; long context does not certify persistent memory.
A_I	I0-IOmega longitudinal persistent-agent datasets, including long-term-memory subtests	Cumulative retention. Memory is embedded: I1 persistence/episodic continuity; I2 consolidation/procedural transfer; I3+ improvement/discovery lineage. No separate A_M is added.
A_O	O0-OOmega organizational datasets	Cumulative role/coordination/evolution tests with external verification.
A_DI	Delivered-correct surface S_A^del(T,H), false-completion/false-rejection outcomes, and intervention logs	Publish F_A(H,p) from delivered-correct reliability. For continuous HLIS, optionally aggregate a predeclared loss-weighted net utility that penalizes false completions (and, if relevant, false rejections).
A_SA	SA0-SAOmega grounded self-model tests	Compare claims about self/state/limitations/change to harness or environment ground truth.

13.5 Why HLIS Uses a Weighted Geometric Mean

A simple arithmetic mean allows an exceptional score in one dimension to compensate too easily for a severe weakness in another. Unified Intelligence is intended to represent a balanced capability envelope, so the continuous pair score should preserve some bottleneck sensitivity even though hard promotion remains governed by U-level gates. The geometric mean supplies this property while remaining smooth and comparable within a declared dimension set.

Profile	Arithmetic mean	Geometric mean	Interpretation
[0.80,0.80,0.80,0.80,0.80]	0.80	0.80	Balanced capability.
[1.00,1.00,1.00,1.00,0.10]	0.82	≈0.63	One severe weakness is visibly penalized.

The geometric mean is not a replacement for the cumulative U gate. A system may have a comparatively high HLIS but remain at a lower U-level because one mandatory prerequisite is not satisfied. Conversely, a system can barely clear a higher U-level while having only moderate continuous strength. Reporting U* and HLIS together preserves both facts.

13.6 The Weight Parameters α_d

For a general leaderboard, equal α_d is the recommended starting point because it minimizes hidden value judgments and keeps systems comparable. Application-specific secondary scores may use different weights, but the weights must be declared before the hidden benchmark is run. They should never be adjusted after observing a model result.

$$\textbf{Equal-weight general score: } \alpha_C = \alpha_I = \alpha_O = \alpha_{DI} = \alpha_{SA} = 1.$$

For example, an AI4Science profile might legitimately emphasize C, I, and DI more than O if the evaluated subject is a single persistent research agent. Such a score should be labeled HLIS_AI4Science and reported beside, not instead of, the general score. This preserves a common headline comparison while allowing application-specific utility estimates.

13.7 Computing Cognitive Achievement A_C

A_C should represent broad cognitive capability rather than raw performance on one saturated benchmark. First normalize benchmark results within declared families, then aggregate within capability domains, and finally aggregate the domains. A possible domain vector is general reasoning, scientific reasoning, mathematics, abstract intelligence, coding, multimodal reasoning, GUI perception/grounding, and long-context cognition. Contemporary benchmarks such as HLE, GPQA, FrontierMath/AIME, ARC-AGI, LiveCodeBench, MMMU/MathVista, ScreenSpot/ScreenSpot-Pro, and RULER/LongBench can contribute evidence, but exact versions and normalization rules must be published. Long-context performance belongs here because it measures use of information made available within an episode; persistent cross-session memory is scored inside A_I instead.

$$A_C = G(R_{general}, R_{science}, R_{math}, R_{abstract}, R_{code}, R_{multi}, R_{GUI}, R_{long}).$$

A domain-balanced geometric or other predeclared aggregation is preferable to pooling all benchmark questions together, because otherwise a domain with many cheap items can dominate a domain represented by fewer but harder tasks.

13.8 Cumulative Achievement for I, O, and SA

Individual Intelligence, Organizational Intelligence, and Self-Awareness are explicitly hierarchical. Let $q_{(d,k)}$ be the externally verified performance on the level-k test suite for dimension d. To prevent level skipping, define cumulative performance at level k by the weakest required lower stage:

$$q^*_{(d,k)} = \min_{j \le k} q_{(d,j)}.$$

A continuous achievement score can then aggregate $q^*_{(d,k)}$ across levels with predeclared level weights w_k:

$$A_d = \left[\sum_k w_k \, q^*_{(d,k)} \right] / \left[\sum_k w_k \right], \quad d \in \{I, O, SA\}.$$

This construction means that a system cannot receive full continuous credit for I4-like behavior if it fails an I2 retention requirement. Higher-level evidence can still provide partial information, but the cumulative envelope remains limited by lower-stage reliability. The same principle implements the paper's central rule $I_{(k+1)} \Rightarrow I_k$, $O_{(k+1)} \Rightarrow O_k$, and $SA_{(k+1)} \Rightarrow SA_k$.

For A_I specifically, memory-specific benchmark performance is no longer folded directly into the I achievement score. Instead, each I-level suite measures the I-specific function - continuity, adaptation, governed self-improvement, recursive improvement, discovery, or open-ended evolution - and the certification rule separately checks M >= mu_n. This prevents a sophisticated memory subsystem from inflating I when the agent does not actually learn or improve, while still blocking an I claim whose memory substrate is insufficient.

Memory receives its own supporting benchmark and may be summarized as an ordinal M-level plus an optional continuous A_M constructed with the same cumulative-retention principle. Recommended M diagnostics include restart persistence, relevant-recall precision/recall, provenance accuracy, temporal reasoning, current-versus-obsolete state, contradiction/update handling, consolidation quality, proceduralization, forgetting/demotion, repair, long-horizon evidence lineage, and downstream usefulness. A_M is reported beside HLIS but is excluded from the default HLIS dimension set D to avoid double counting with I.

Formal I certification rule: certify I_n only if q*_(I,n) reaches its predeclared threshold AND M(A) >= mu_n. Memory can exceed this prerequisite without promoting I; therefore M>=mu_n is necessary but not sufficient.

13.9 Delegation Achievement A_DI: Capability, Delivery Integrity, and Human Load

Delegation Intelligence has three related but non-identical observables: task capability, delivery integrity, and human load. The certification frontier is based on delivered-and-correct completion S_A^del(T,H). Outcome integrity is measured by false completion and false rejection. Human cognitive load remains represented by H, Critical Intervention Depth, intervention count/minutes, and any escalation required after held-back work.

$$\mathbf{F\_A(H,p) = \max \{ T : S\_A(T,H) \geq p \}.}$$

13.9.1 Why a success-only aggregate is insufficient

If two paired harnesses differ only by a post-hoc acceptance step, the external verifier verdict on the produced artifact is unchanged. Therefore a gross score defined only as P(V=1) is invariant to acceptance even when false completions fall sharply. This is not a defect in verifier accuracy; it is a mismatch between the scoring variable and the delegator-facing outcome.

Proposition 1 (paired acceptance invariance). For the same executor artifact, adding a purely post-hoc accept/hold-back step cannot change P(V=1). Any gross delegation aggregate built only from verifier-pass probability is therefore identical across those paired rungs. A benchmark that wants to value outcome integrity must record delivery/acceptance separately.

13.9.2 Outcome-complete delegation scoring

For every attempt record D (delivered/accepted) and V (external verifier pass). This yields four outcomes: delivered-correct (D=1,V=1); false completion (D=1,V=0); held-back invalid work (D=0,V=0); and false rejection (D=0,V=1). The first is successful delegation; the second is the most important reliability failure; the third is a non-completion that should appear in human-load/escalation metrics; and the fourth measures excessive strictness.

A loss-weighted net surface may be used as a secondary continuous scoring primitive:

S_A^net(T,H; lambda_FC, lambda_FR) = P(D=1,V=1) - lambda_FC P(D=1,V=0) - lambda_FR P(D=0,V=1).

The loss multipliers lambda_FC and lambda_FR are application/task-class parameters and must be fixed before evaluation. If false rejection has negligible cost, lambda_FR may be zero. Unlike the earlier success-only aggregate, the positive term is delivered-and-correct rather than verifier-pass, so a correct result that the harness refuses to deliver is not credited as successful delegation.

A_DI = [sum_(t,h) w_t v_h clamp(S_A^net(T_t,H_h),0,1)] / [sum_(t,h) w_t v_h].

Task weights w_t increase with difficulty and intervention weights v_h increase as required human cognition decreases. One illustrative intervention schedule is H0=6, H1=5, H2=4, H3=3, H4=2, H5=1; all such weights and loss ratios are calibration choices, not laws. The scalar A_DI must never replace the frontier: at minimum publish F_A(H0,p), F_A(H1,p), F_A(H2,p), false-completion rate, false-rejection rate, and human-load statistics.

13.10 Numerical Example of HLIS

Consider an organizational harness-LLM pair with equal weights and normalized achievements A_C=0.90, A_I=0.70, A_O=0.60, A_DI=0.80, and A_SA=0.75. Then:

HLIS = 100 × (0.90 × 0.70 × 0.60 × 0.80 × 0.75)^(1/5) ≈ 74.3.

Dimension	Achievement
C	0.90
I	0.70
O	0.60
DI	0.80
SA	0.75

The recommended report is not merely “HLIS=74.3.” It should preserve the discrete profile and certification context, for example: U3 | HLIS 74.3 | [C5,I3,O3,T4/H1,SA3] | p=0.80 | false-completion rate=0.00 | benchmark version Bv1.5 | ladder id=L-HIL-1 | resource envelope R-Standard. This allows a reader to see both the continuous aggregate and the source of the score.

13.11 Zero Scores, N/A Dimensions, and Hard Gates

A zero in a required geometric-mean component forces HLIS to zero. This can be appropriate if the system completely fails a foundational required dimension, but it must not be confused with “not applicable.” For a singleton system, O is omitted from D rather than set to zero. If a benchmark produces a true zero but a continuous nonzero pair score is analytically desired, a small preregistered floor ε may be used only as a secondary sensitivity analysis; the primary U-level gate should remain strict. This paper therefore recommends hard ordinal gates plus a soft continuous HLIS, not replacing the gate with numerical smoothing.

13.12 Run Logs Must Record Acceptance and Delivery Outcomes

A run log that stores only verifier_pass cannot distinguish a wrong artifact that was confidently delivered from one that an acceptance mechanism correctly held back. Formal HIL runs should therefore record verifier_pass and harness_accepted/delivered separately. For harnesses without an acceptance step, harness_accepted is recorded as null and delivery is treated as true by convention.

Required field	Meaning
verifier_pass	External ground-truth/check verdict on the artifact.
harness_accepted / delivered	Whether the tested harness returned the artifact as completed work.
false_completion	Derived: delivered=1 and verifier_pass=0.
held_back_invalid	Derived: delivered=0 and verifier_pass=0.
false_rejection	Derived: delivered=0 and verifier_pass=1.
attempts	Number of executor attempts consumed.
verification_responsive	Whether a verifier rejection was converted into a later verified correction without human help deeper than CID1.

13.13 Reliability, Confidence Intervals, and Resource Envelopes

Every achievement estimate is sampled from tasks and therefore uncertain. HLIS should eventually be accompanied by a confidence interval, for example HLIS=74.3±1.8 or a bootstrap 95% interval. The same benchmark tasks should be rerun across seeds or matched task families when stochasticity is material. A one-off success at a high level is not equivalent to reliable capability.

Resource conditions must also accompany the score. The complete notation HLIS(m,h;D,α,R,B) makes explicit that a 500-call multi-agent harness and a one-call harness are not automatically comparable if their resource envelopes differ. Recommended R fields include total model tokens, number of LLM calls, wall-clock time, compute class, persistent-memory budget, tool/API access, external knowledge access, and number of subagents. Cost and latency should be reported beside intelligence rather than silently included in it. A separate efficiency or Pareto analysis can compare intelligence against cost and time.

13.14 HLIS Is the Building Block of HIL

HLIS evaluates one point: one frozen LLM under one harness. HIL evaluates the same frozen LLM across a standardized ladder of harness generations. If HG_k is the kth standardized harness, the Harness Scaling Curve is:

HSC_m(k) = HLIS(m, HG_k).

The sequence {HLIS(m,HG0), …, HLIS(m,HGK)} is then used to compute HIL-Level, HIL-AUC, HIL-Ceiling, and Harness Gain. In this sense, HLIS is to one harness-LLM system what an individual point is to a scaling curve; HIL characterizes the model’s ability to convert increasingly capable harness structure into verified intelligence.

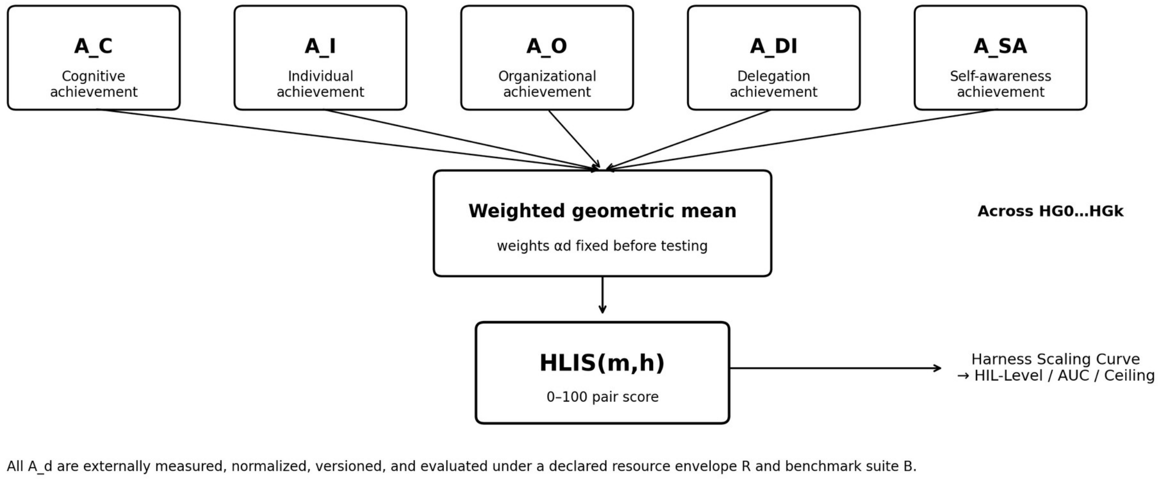

Figure 9. HLIS aggregates externally measured intelligence-family achievements for one LLM-harness pair. Repeating the pair evaluation across a standardized harness ladder produces the Harness Scaling Curve and the higher-level HIL characterization.

13.15 Recommended Pair-Level Reporting Standard

Field	Example
System	Model-X + HG3-v1.0
Unified level	U3
HLIS general	74.3 [95% CI 72.7-75.9]
Core profile	C5 | I3 | O3 | DI frontier H0:T3, H1:T4 | SA3
Reliability	p*=0.80 certification frontier
Dimension set / weights	D={C,I,O,DI,SA}; equal α_d
Benchmark suite	Bv1.3; hidden verifier set frozen before test
Resource envelope	R-Standard: declared tokens/calls/time/tools/subagents
Efficiency context	Cost/task and latency/task reported separately
Per-coordinate headroom	Fraction of items not maxed by strongest available executor; saturated coordinates explicitly marked.
Item-validity audit	Last cross-tier concordance audit date; items flagged; resolution status.
Specification-key consistency	Pass/fail for parameterized families; test executes the key reference implementation.
Graded-item distribution	Publish distribution and separated failure modes, not only aggregate mean.
Supporting memory	M4 | A_M=0.82 (example); not included in HLIS product

This reporting standard prevents HLIS from becoming an opaque single number. A pair report is incomplete unless it carries U*, HLIS with dimension set and weights, the discrete profile, the named bottleneck, reliability with an interval, the delegation frontier, false-completion and false-rejection rates, human intervention/CID, the resource envelope, benchmark version, and ladder identifier. Version 1.7 additionally requires per-coordinate headroom, the date/status of the last cross-tier concordance audit, specification-key consistency status for parameterized item families, and score distributions for graded item families. A coordinate with near-zero headroom is reporting the benchmark ceiling, not an established model ceiling.

14. Harness-Intelligence-Level (HIL) Characterization of a Base LLM

14.1 Why an LLM Needs a Harness-Based Score

A base LLM should not be judged only in a minimal chat harness, because some models may be unusually capable of using memory, planning, tools, multiple roles, verification, and self-improvement machinery. Conversely, an elaborate harness can mask a model that contributes little. To measure the ability of one LLM to express intelligence through harnesses, evaluate the same frozen model across a standardized harness ladder H = {HG0, HG1, ..., HG6, HGOmega}. Tools, external knowledge access, resource ceilings, authority, and hidden benchmarks should be standardized so the comparison reflects model-harness interaction rather than uncontrolled infrastructure advantages.

14.2 HIL-Level, HIL-AUC, HIL-Ceiling, and Harness Gain

Metric	Definition	Interpretation
HIL-Level(m)	Highest standardized harness generation k for which A(m,HGk) satisfies the corresponding cumulative target gate.	How far up the harness ladder the LLM can validly exploit the standardized architecture.
HIL-AUC(m)	Weighted mean of HLIS(m,HGk) across all tested standardized harness generations.	Broad harness-expressible ability; rewards models that work well across multiple harness complexities.
HIL-Ceiling(m)	max_k HLIS(m,HGk).	Best intelligence score that a standardized harness can realize from the model.
Harness Gain(m)	max_k HLIS(m,HGk) - HLIS(m,HG0).	How much capability is unlocked by harnessing beyond the minimal baseline.
Harness Design Score (HDS)	Mean independently verified gain produced when the LLM proposes candidate harness improvements under a fixed design budget.	Ability of the LLM to improve the architecture that harnesses it; reported separately from baseline HIL.
Verification Responsiveness VR	Fraction of verifier-rejected attempts converted into a later verified accepted correction without human help deeper than CID1.	How effectively the model-harness pair uses external failure evidence; feedback protocol must be standardized.

14.3 Verification Responsiveness VR

Harness Gain says how much a richer harness helped, but not which part required the model. Acceptance and rollback can reduce harmful delivery even if the model contributes nothing after its first attempt. Retry, replanning, and routing require the executor to convert detected failure into a correction. Verification Responsiveness measures that interaction directly.

VR(m,h) = (# verifier-rejected attempts followed by a verified accepted correction with no human help deeper than CID1) / (# verifier-rejected attempts eligible for retry).

VR must be reported with the granularity and correctness of the rejection feedback. A model cannot respond to a misleading or underspecified failure message; therefore VR is a pair-level interaction diagnostic with a model contribution, not a pure intrinsic-model scalar. It should be measured under standardized feedback protocols when comparing LLMs.

14.4 A Single HIL Score and Structural Cautions

A single HIL-Score may be published as a secondary convenience, but Version 1.5 treats the Harness Scaling Curve and a compact summary tuple as primary. HIL-AUC and HIL-Ceiling are structurally correlated because Ceiling is the maximum of the same values that AUC averages; moreover, AUC can rank a high-baseline flat model above a lower-baseline steep model even when harness response is the property of interest. Any composite weights therefore require empirical calibration against downstream outcomes rather than aesthetic choice.

Recommended primary summary: <HIL-Level, HLIS(HG0), HIL-Ceiling, Harness Gain, VR, ladder id>, with the complete Harness Scaling Curve published beside it. If the earlier illustrative composite HIL-Score = 0.55*AUC + 0.35*Ceiling + 0.10*Harnessability is retained for continuity, label it provisional and publish every component.

A high HIL-Score therefore means more than "the model answers hard prompts." It means that, across a fixed suite of increasingly capable harnesses, the LLM repeatedly converts architectural support into externally verified intelligence. This makes HIL a candidate metric for comparing the harness-expressible ability of different LLMs.

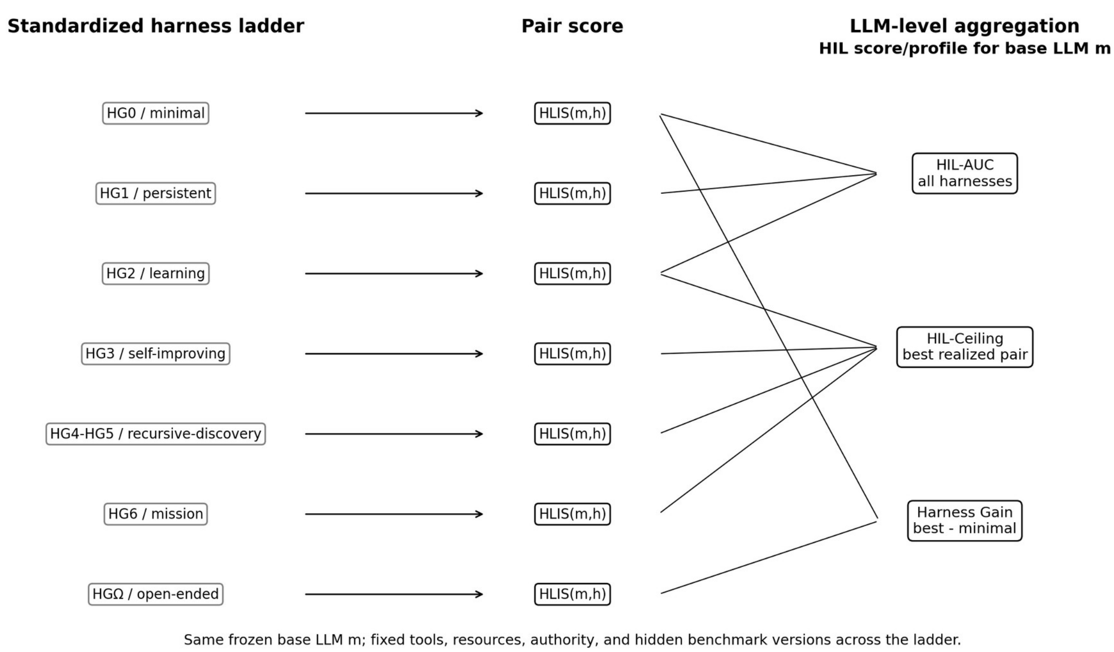

Figure 6. The same base LLM is tested across a standardized harness ladder. Each pair receives HLIS; the collection of pair scores yields HIL-Level, HIL-AUC, HIL-Ceiling, Harness Gain, and an optional composite HIL-Score.

14.5 Recommended LLM Scorecard

Score	Subject being measured	Primary use
C-Score / model baseline	Base LLM in a minimal standardized harness.	Raw/broad cognitive problem-solving comparison.
Delegation Surface / DI score	One LLM-harness pair.	How hard a task can be delegated at each human-intervention budget.
Unified level U*	One LLM-harness pair or organization.	Highest cumulative intelligence stage actually certified.

Score	Subject being measured	Primary use
HLIS	One frozen LLM-harness pair.	Continuous 0-100 comparison within/across levels.
HIL-Level + Harness Scaling Curve	One base LLM across the standardized harness ladder.	How much intelligence the LLM can express across standardized harnesses; report baseline, ceiling, gain, VR, and optional composite.
HDS / Harness Design Gain	LLM acting as a harness designer under frozen external evaluation.	How well the model can design architectures that improve its own or another model's realized intelligence.
Verification Responsiveness VR	One base LLM measured through standardized retry/feedback harnesses.	Whether detected failure is converted into verified correction; separates passive acceptance gains from model-responsive harness gains.

15. Benchmark Library for Harness-Level Intelligence

A harness intelligence program should maintain a versioned benchmark registry rather than one monolithic test set. Every core coordinate level has a harness contract and matched dataset, and Version 1.8 adds M-Bench as a separate supporting benchmark. The minimal unit is a task manifest containing target coordinate/level, frozen model and harness hashes, tool/resource/authority budget, intervention policy, hidden verifier, success threshold, and required retention/prerequisite suites.

15.1 Task Binding Status

A registry of task specifications is not automatically an executable benchmark. Every task row should record one of three statuses: bound (a runnable instance and verifier exist for this family/band); band-only (some runnable task exists at the same difficulty band but not this family); or specification-only (the task is defined but not instantiated). Band-only must not be counted as coverage, because substituting another domain at the same T band changes the construct being measured.

M-Bench should independently cover: restart persistence; relevant retrieval under distractor load; episodic provenance; temporal ordering and current-versus-obsolete state; knowledge updates and contradiction resolution; semantic consolidation; proceduralization; forgetting/demotion; memory-health detection and repair; long-horizon research/decision lineage; and, at MΩ, externally evaluated changes to memory representations/retrieval/consolidation mechanisms. I-Bench consumes M-Bench as a prerequisite but scores I-specific behavior rather than re-scoring these same memory functions.

Suite	Purpose	Example level-specific evidence
DLI-Bench	First-stage delegation surface.	T0 atomic execution; T2 multi-step persistence; T3 strategy construction; T4 long-horizon project; T5 hidden solution; T6 mission; Tomega charter cycles, each under H budgets.
U-Bench-U0...UOmega	Unified promotion and lower-level retention.	At Un, rerun U0...U(n-1) retention plus the new gate for C/I/O/DI/SA.
I-Bench-I0...IΩ	Persistent individual evolution, scored separately from memory.	I1 continuity/resumption; I2 verified adaptation/transfer; I3 governed cognitive self-improvement; I4 meta-improvement; I5 discovery; IΩ open-ended individual evolution. Each level declares a minimum M prerequisite.

Suite	Purpose	Example level-specific evidence
M-Bench-M0...MΩ	Independent supporting Memory Capability.	Persistence; structured episodic/provenance/temporal memory; consolidation/proceduralization; self-management/repair; longitudinal knowledge/research lineage; optional externally evaluated memory-mechanism evolution.
O-Bench-O0...OOmega	Collective intelligence.	Routing, persistent organization, adaptive coordination, reorganization, meta-improvement, collective discovery, new-role creation.
SA-Bench-SA0...SAOmega	Evidence-grounded self-awareness.	State truth, capability limits, causal self-diagnosis, predicted self-change, self-unknowns, mission identity, lineage.
C-Bench	Broad cognitive capability.	Multiple reasoning/domain families to avoid mistaking one specialty for general cognition; long-context handling is a C subdomain, not an I1 memory certification.
Circle-Bench	Application/substrate reach, reported orthogonally.	Software closure first; later hardware/physical/biological/other substrate-specific implementation and validation loops.
GUI-Bench / GP panel	Diagnostic computer-screen perception and grounding inside C.	GP0-GP5 static/dynamic screenshot tasks; freeze resolution and observation channel; report grounding, UI-state and screen-change errors. End-to-end computer use is scored in DI.

This organization supports the proposed development workflow: start with Delegation Intelligence because it is immediately operational; then certify a Unified level; then add the coordinate suites most important for the target application. A research worker might emphasize I and SA, a fleet O and DI, an LLM benchmark C and HIL, and a physical autonomy program DI plus Circle completion. The common scores remain comparable because the application panels are reported in addition to, not instead of, the standardized core.

15.2 GUI Perception and Computer-Use Benchmarking

Computer-agent evaluation requires two linked but distinct suites. GUI-Bench/GP isolates whether the model can correctly perceive and ground the screen; DI computer-use tasks test whether the full model-harness system can turn that perception into correct closed-loop work with bounded human cognition. A single end-to-end success rate cannot tell these failure modes apart.

Layer	Primary question	Representative evidence	HIL interpretation
GP / C	Can the system see and understand the interface?	ScreenSpot; ScreenSpot-Pro; HIL-native screenshot/state tests	C subscore and diagnostic profile; no new top-level HLIS dimension.
Task P_vis	How visually demanding is the task?	Screen density, resolution, target size, windows, overlays, dynamic change	Task attribute used to stratify difficulty and compare like with like.
DI computer use	Can the system complete the computer task with bounded human cognition?	OSWorld-family executable tasks; HIL-native desktop workflows	S_A^del(T,H), frontier F(H,p), false completions/rejections, CID and human minutes.
Closed-loop diagnosis	Where did failure occur?	[P,G,S,A,R] pipeline and per-step logs	Separates perception/grounding/state errors from planning, action

Layer	Primary question	Representative evidence	HIL interpretation
			and verification failures.

For HIL comparisons, computer-agent observation channels are part of the harness manifest. Raw screenshots, OCR, accessibility trees, set-of-mark overlays, element detectors and persistent screen-state trackers are different harnesses, not interchangeable benchmark settings. A valid Harness Scaling Curve therefore freezes or explicitly ladders these mechanisms rather than mixing them across models.

16. The Validity of the Benchmark Items

Everything in HLIS and HIL is arithmetic over benchmark item outcomes. An item with a hidden answer key is therefore part of the measuring instrument, and the score inherits any defect in that item. Version 1.6 demonstrated this directly: two hidden-key defects had passed their own test suite and had been scored as model failures. The framework therefore requires explicit item-validity controls before a failure is attributed to an executor.

16.1 Trickiness Does Not Desaturate a Suite; Difficulty Does

The first attempt to create headroom used three tasks built around known reasoning hazards: causal versus timestamp order, daily aggregation across a daylight-saving transition, and Unicode identity normalization. The tasks perfectly separated a scripted-correct implementation from a naive one, yet capable live executors all passed. The lesson is that a hazard that must be disclosed to make the item well-posed becomes a reading-comprehension requirement rather than durable difficulty.

Item family	Scripted-correct solver	Naive solver	Live executors
Three disclosed-hazard DL3 items	10/10	0/10	All probes passed across both tested tiers

Difficulty that survives full disclosure should instead come from scale, long interacting specifications, search/optimality, horizon, dynamic state, multimodal grounding, or other demands that remain difficult even when the rules are explicit. This principle is especially relevant to GUI tasks: hiding an icon label is trickiness; requiring precise grounding across a dense dynamic professional interface is perceptual difficulty.

16.2 Two Harder Item Families

Version 1.6 built two DL4 families on two such axes. The first increased specification scale with a small expression language containing interacting syntax, precedence, truthiness, typing and dialect rules. The second required exact minimum-cost covering under explicit constraints, so feasibility and optimality were separate questions. Both families were parameterized so scored instances could be hidden without hiding the rules.

Family	Difficulty axis	Scoring unit	Separated failure modes
Expression language	Specification scale / interacting rules	Fraction of generated expressions interpreted correctly	Rule/precedence/binding/type failure classes
Minimum-cost covering	Search / exact optimality	Feasibility plus exact optimum across generated instances	Infeasible; feasible-but-suboptimal; crash; timeout; not attempted

The covering family illustrates why boolean scoring is insufficient: a greedy solver can be feasible on every case while being rarely optimal. A single failed bit hides whether the executor misunderstood the specification, searched inadequately, crashed, timed out, or never attempted the task.

16.3 Cross-Tier Concordance Auditing

When the expression family was run on executors of materially different capability, both returned the same wrong value on the same cases while the hidden key disagreed. That pattern is evidence about the item, not merely about the models. Two defects were found: one varying dialect rule was graded differently from what the visible specification stated, and one binding consequence was under-specified. After repair, the suspicious cases disappeared.

Seed	Frontier executor	Weaker executor	Audit reading
0	0.941	0.933	Both wrong on same case -> item audit triggered
1	0.958	0.958	Both wrong on same case -> item audit triggered
2	1.000	0.996	No concordant-key defect signal
3	1.000	1.000	No failure

Requirement: before identical failures are attributed to executors, any item on which materially different executors return the same response and the key disagrees must undergo cross-tier concordance audit. The audit does not replace expert review; it supplies an independent signal precisely when the asserted ground truth is what is in doubt.

16.4 Specification-Key Consistency in Parameterized Item Families

Parameterized families introduce a failure mode that fixed items do not: a rule can vary in the key without varying in the visible prose, or vice versa. Every varying rule must therefore ship with a test that samples multiple generated instances, extracts the values claimed by the specification, and verifies that the key's own reference implementation returns those exact values. The test must exercise the key implementation itself; two separately written implementations of the same misunderstanding can agree and still be wrong.

The same principle applies to stated constraints. If an item promises a time limit, action budget, observation channel, or GUI resolution, the scorer must enforce it. A requirement that appears only in prose is not part of the measured task.

16.5 Graded Item Scoring and Separated Failure Modes

Where an item contains many subcases or rules, graded scoring should preserve partial information rather than collapsing all performance to pass/fail. A system that correctly handles 99% of a 238-case generated family is not in the same diagnostic state as one that returns an empty artifact, even if both miss a strict perfect-score threshold. The distribution of grades must be published beside the mean, and failure modes should be separated when they imply different repairs.

For GUI benchmarks, the same rule means reporting at least element recognition/grounding accuracy, interface-state errors, action-target errors, and result-change recognition rather than only final task success. End-to-end DI success remains necessary, but the item-level decomposition shows which subsystem limited it.

16.6 Headroom and What the Corrected Items Measured

After the two expression-item defects were repaired, the family separated the weaker executor from the frontier executor with one interpretable recurring failure mode, but the frontier executor scored 1.000 on every seed. The covering family also failed to desaturate the frontier. This is an important negative result: a benchmark may discriminate some systems while still providing no evidence about the strongest system's ceiling.

Family / executor	Observed result	Interpretation
Expression / frontier	6/6 seeds; mean 1.000	Coordinate still saturated at frontier
Expression / weaker	2/6 perfect; mean 0.993; 0.975-1.000 range	Small but interpretable nested-binding failure mode
Covering / frontier	4/4 seeds; mean 1.000	Exact optimizer implemented; no frontier headroom
Covering / weaker	3 scored seeds at 1.000; one unmodified stub delivery	Distinguish not-attempted/false-completion from search capability

Per-coordinate headroom is therefore part of the HIL report: the fraction of items on which the strongest available executor is not already at the maximum score. A coordinate with near-zero headroom is bounded by the suite. Its HLIS contribution and any Harness Scaling Curve segment must carry a saturation statement rather than being interpreted as an established model ceiling.

17. Relationship to Contemporary Model and Agent Benchmarks

17.1 Existing Benchmarks Measure Important Slices, Not a Single Intelligence Quantity

Modern model evaluation is intentionally plural. HLE and MMLU-Pro emphasize broad difficult academic reasoning; GPQA targets graduate-level scientific questions; FrontierMath and competition mathematics target advanced mathematical reasoning; LiveCodeBench and SWE-bench probe coding and repository-level software work; ARC-AGI emphasizes fluid adaptation on novel tasks; BFCL evaluates tool/function calling; the tau-bench family evaluates interactive tool-agent-user reliability; BrowseComp and GAIA probe browsing and general assistant behavior; MMMU and MathVista evaluate multimodal reasoning; RULER and LongBench probe long-context behavior; SimpleQA targets factuality; Chatbot Arena measures human preference; and OSWorld evaluates multimodal computer-use agents [3-20]. These benchmarks are complementary because they deliberately stress different capabilities. LongMemEval and LoCoMo explicitly probe long-term conversational memory, while LongMemEval-V2 extends the question toward agent experience in specialized environments [21-24]. These are especially relevant to the I1-I2 evidence layer, but they still do not by themselves establish governed self-improvement or recursive evolution.

HIL therefore should not be presented as replacing these scores. Let B(m) denote the vector of contemporary benchmark results for base model m. The HIL program consumes B(m) as part of the evidence layer, then asks a different question: how does the same frozen model behave when embedded in progressively stronger, standardized harness architectures? The central distinction is benchmark slice versus harness-expressible system intelligence.

17.2 Mapping Contemporary Benchmarks into the HIL Measurement Architecture

Capability slice	Representative benchmark families	Primary HIL evidence	What the benchmark alone does not establish
Broad reasoning	HLE; MMLU-Pro	C baseline / breadth and hard closed-ended reasoning	Persistence, autonomy, organization, self-improvement, or harness scaling.
Scientific reasoning	GPQA Diamond	C-science; expert scientific reasoning evidence	Longitudinal scientific learning or autonomous research execution.
Advanced mathematics	AIME-style competition math; FrontierMath	C-math; difficult and frontier mathematical reasoning	Project autonomy, tool orchestration, or learning across episodes.
Coding	LiveCodeBench	C-code plus bounded execution evidence	Repository-scale planning, persistence, human intervention, or organization.
Software engineering	SWE-bench and newer repo-level suites	DI at T2-T4; C-code; execution/verifier evidence	A general intelligence level unless human intervention and cross-domain retention are controlled.
Abstract / adaptive intelligence	ARC-AGI-2/3 family	C-fluid; DI exploration/adaptation; efficiency evidence	Persistent individual evolution, organizational evolution, or broad real-world coverage.
Tool calling	BFCL	DI tool-selection/action layer; abstention and multi-step evidence	Long-horizon mission autonomy by itself.
Interactive agent reliability	tau-bench family and maintained descendants	DI success surface; reliability p; policy/tool interaction	General cognition or self-improvement depth.
Research / browser agent	BrowseComp; GAIA	DI for research tasks; C retrieval/reasoning; tool reach	Longitudinal learning, organizational evolution, or self-model depth.
Vision + reasoning	MMMU; MathVista; ScreenSpot; ScreenSpot-Pro	C-multimodal plus GP perception/grounding diagnostics	End-to-end computer task completion, persistence, or harness scaling.
Long context	RULER; LongBench	C-context: retrieval and reasoning over information present within the evaluation episode	Persistent cross-session memory, restart recovery, consolidation, provenance, or useful learning over time; those belong to I.
Factuality	SimpleQA and related factuality suites	C-factuality / calibration evidence	Autonomy, organization, or self-improvement.
Real-user preference	Chatbot Arena / human preference	External preference / usability evidence; useful secondary panel	A mechanistic or cumulative intelligence level.
Computer agent	OSWorld family; HIL-native desktop workflows	DI in executable GUI environments; GP/C diagnostics; delivered-correct task completion	Longitudinal I/O/SA maturity unless separately instrumented; observation-channel effects unless frozen.
Long-term memory / experience	LongMemEval; LoCoMo; LongMemEval-V2; memory-harness evaluations such as MemGPT-style multi-session settings	I1 persistence/episodic retrieval; portions of I2 temporal reasoning, knowledge update, and experience use	Higher I2 consolidation/procedural learning, I3+ self-improvement lineage, or Unified level unless separately tested.

Capability slice	Representative benchmark families	Primary HIL evidence	What the benchmark alone does not establish
GUI grounding / screen perception	ScreenSpot; ScreenSpot-Pro	GP ⊂ C: element recognition, spatial grounding, interface state and screen-change understanding	Planning, autonomous task completion, memory, or organization by itself.

The mappings are deliberately many-to-many. A benchmark can contribute evidence to more than one coordinate, but no benchmark is allowed to certify an unrelated coordinate merely because it is difficult. Reports must name the exact benchmark version, task split, harness, tool/resource budget, and evaluation date because benchmark families evolve over time. For example, BFCL and the tau-bench line have evolved across versions and increasingly agentic settings; HIL should therefore cite frozen benchmark versions rather than benchmark names alone [10-11].

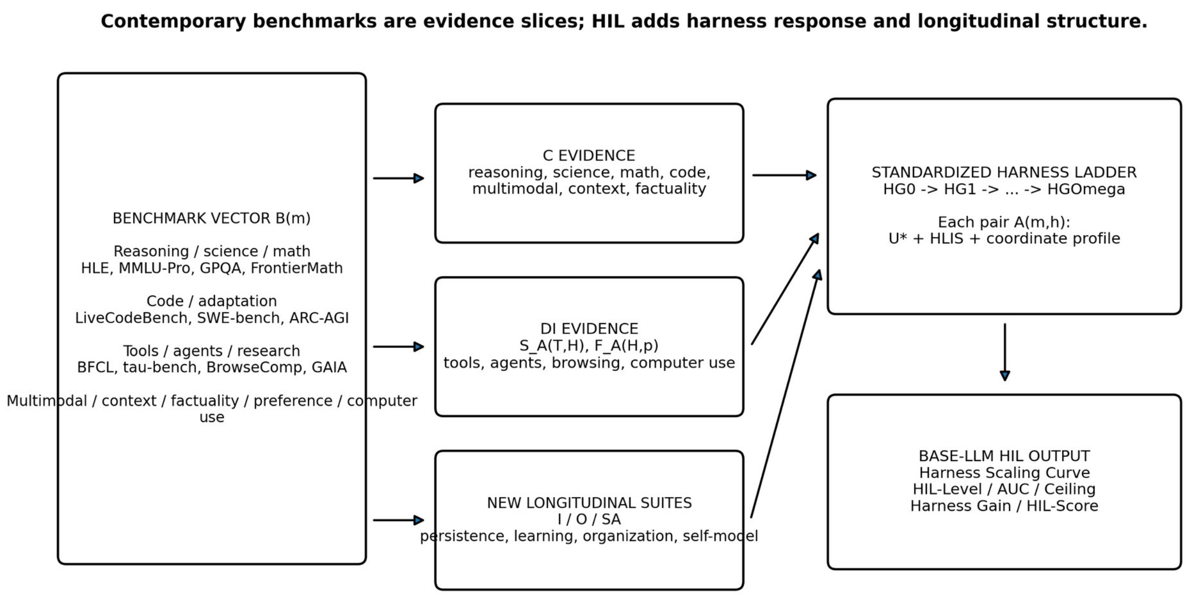

Figure 7. Contemporary benchmarks remain the evidence layer. HIL organizes their evidence into cognitive, delegation, and longitudinal/structural coordinates, then measures how those capabilities compose under a standardized harness ladder.

17.3 The Main Added Quantity: Harness Response

Most benchmark leaderboards primarily compare systems at one evaluation configuration. HIL adds an explicit response-to-harness dimension. For frozen model m and standardized harness generation HG_k, define HSC_m(k) = HLIS(m, HG_k). The ordered set HSC_m = {(k, HLIS(m,HG_k))} is the Harness Scaling Curve. It reveals whether model capability continues to convert into higher realized intelligence as persistence, tools, delegation, learning, organization, verification, and self-improvement machinery are added.

Two models can therefore reverse order as harness complexity increases. A model with a stronger minimal-harness benchmark profile may plateau early, while another with a weaker HG0 score may exploit persistent long-term memory, planning, verifier feedback, multi-agent protocols, and long-horizon state much more effectively. HIL-AUC, HIL-Ceiling, Harness Gain, and HIL-Level summarize this curve, but the curve itself should remain a primary result because it exposes saturation and non-monotonic behavior.

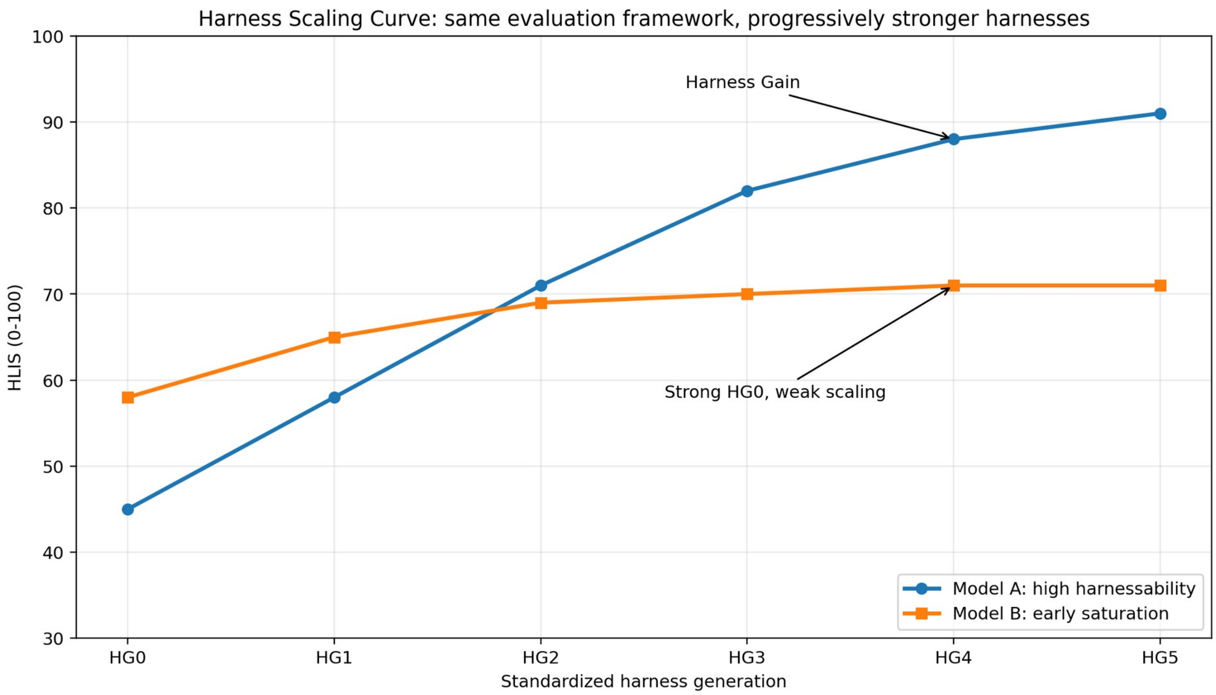

Figure 8. Illustrative Harness Scaling Curves. Model B is stronger in the minimal harness but saturates; Model A converts stronger harness generations into substantially higher realized intelligence. The curves are hypothetical and illustrate the measurement concept, not empirical model results.

17.4 Pilot Reference Measurement: What Building the Ladder Revealed

A preliminary reference implementation built HG0-HG3 for Delegation Intelligence and ran three executors from two vendor families under a fixed H1 intervention budget. The study is intentionally small and is not presented as a product ranking. Its value is methodological: it exposed a scoring invariance and several experimental-design requirements that were not obvious from the equations alone.

17.4.1 Paired acceptance invariance

For paired HG0/HG1 outputs, gross verifier success is exactly identical by construction because the executor artifact is the same. A separate acceptance process can nevertheless convert false completions into held-back invalid work. In the controlled paired check, gross success stayed fixed while the net outcome score improved and false rejections were zero. This is the empirical motivation for separating verifier verdict from delivery/acceptance in Section 13.9.

Executor	**Gross HG0**	**Gross HG1**	**Net HG0**	**Net HG1**	**False rejections**
Family A, stronger	0.933	0.933	0.867	0.933	0
Family A, weaker	0.917	0.917	0.833	0.917	0

17.4.2 Harness response, saturation, and cost

Across the initial four-rung pilot, stronger frontier executors had little headroom and produced nearly identical aggregate curves, while a weaker executor gained more from retries and harness support. Identical aggregate curves did not imply equivalent executors: they failed different instances and differed materially in wall-clock cost. The lesson is that a Harness Scaling Curve is only as discriminative as its task-suite headroom, and intelligence scores must be reported beside resource envelopes and per-instance failure patterns.

17.4.3 Paired-rung protocol

Whenever two rungs do not alter the executor request and differ only in post-hoc acceptance or evaluation machinery, run the executor once per task/seed and score both rungs from the same artifacts. Independent reruns introduce stochastic variance that can be larger than the mechanism under study. The paired design is both cleaner and cheaper.

17.5 Why HIL Adds Information Beyond a Benchmark Average

Property	Typical benchmark score	HIL extension
Unit of analysis	Model or agent in one benchmark configuration.	Frozen LLM-harness pair A(m,h), plus the same LLM across a harness ladder.
Capability breadth	A vector of specialized scores.	Preserves the vector but organizes it into C/DI/application evidence and adds I/O/SA longitudinal tests.
Autonomy	Sometimes implicit or benchmark-specific.	Explicit T x H x p delegation surface and frontier F_A(H,p).
Human dependence	Often not a first-class reported variable.	H0-H5 cognitive intervention is logged and constrained.
Persistence / learning	Usually one episode or fixed model state.	I-level tests embed long-term memory: real restart continuity, episodic retrieval/provenance, temporal update handling, consolidation/procedural transfer, and later governed self-improvement lineage.
Organization	Often a fixed agent architecture.	O-level tests evaluate coordination, reorganization, and collective improvement with real role boundaries.
Self-awareness	Usually absent or conversational.	SA requires evidence-grounded state, limitation, causal-failure, and change-lineage tests.
Model versus harness progress	Usually conflated in the final system score.	HLIS measures one pair; Harness Gain and HIL separate model baseline from harness unlock.
Higher-order improvement	Not normally measured.	HG3+ can test whether the system improves cognition/organization under frozen external evaluation.

The strongest claim for HIL is therefore not that it produces one universally superior scalar. Its value is that it measures a new response function: how much externally verified intelligence a base LLM can express as the harness around it becomes more capable. The overall HIL-Score is a convenience summary; the benchmark vector B(m), Delegation Surface, Unified level, coordinate profile, and Harness Scaling Curve remain the primary scientific outputs.

17.6 Recommended Two-Layer Scorecard for Frontier LLMs

Layer	Report	Purpose
Contemporary benchmark layer	HLE/MMLU-Pro; GPQA; math; coding/SWE; ARC-AGI; BFCL; agent/research; multimodal; long context; factuality; human preference; computer-use benchmarks, with exact versions.	Shows the familiar capability slices and allows comparison with existing leaderboards.
HIL layer	C profile; S_A^del(T,H), F_A(H,p), false-completion/rejection rates; I/O/SA; U*; per-harness HLIS; HSC; HIL-Level; baseline; Ceiling; Gain; VR; optional calibrated composite.	Shows how the model becomes a persistent, delegated, organizational, self-modeling system as harness sophistication increases.

This two-layer reporting strategy avoids the central failure mode of a single leaderboard number. A model can remain visibly strong in science but weaker in coding or tool use, while HIL separately reveals whether those abilities compose into reliable autonomous systems. For real deployments, the application-specific coordinate panel from Section 12.3 is then added without altering either layer.

17.7 Relationship to Prior AGI-Level Proposals

The proposed hierarchy is related to, but distinct from, prior AGI taxonomies. Morris et al. separate performance, generality, and autonomy and argue for staged operational definitions rather than a binary AGI label [1]. Hendrycks et al. define AGI in terms of the versatility and proficiency of a well-educated adult across a psychometrically grounded set of cognitive domains [2]. The present framework preserves broad cognitive evidence in C, but adds persistent individual evolution I, organizational evolution O, operational self-awareness SA, the explicit Delegation Surface (T,H,p), and a standardized harness-response characterization. HIL is therefore best viewed as a harness-centric complement to capability- and human-reference-centered AGI definitions rather than a replacement for them.

18. Benchmark and Promotion Protocol

- Freeze the tested model, harness, prompts, tool set, authority profile, resource envelope, benchmark version, and ladder identifier before certification.
- When two rungs differ only in post-hoc machinery, use the same executor artifacts for both rungs; do not introduce unnecessary between-rung stochasticity.
- Record verifier_pass and harness_accepted/delivered separately so false completions, held-back invalid work, and false rejections remain observable.
- Publish task binding status (bound, band-only, specification-only) and do not count band-only tasks as coverage for an uninstantiated family.
- For harness-design experiments, the designer model may not alter or select the hidden certification set, verifier, or promotion criterion.
- Freeze the tested harness, model, tool set, prompts, and authority profile before certification.
- Use multiple task families per T band so difficulty is not synonymous with one domain such as software.
- Run at predeclared intervention budgets H0-H5 with a written human-assistance policy.
- Use an independent verifier or sealed reference; the performing agent cannot certify its own success.
- Record cognitive interventions separately from governance approvals.

- Record Critical Intervention Depth so a single human-supplied central strategy cannot be hidden by a low intervention count.
- Use confidence intervals and repeated tasks to estimate S_A(T,H), then report the frontier F_A(H,p).
- For U promotion, rerun a retention suite for all lower levels and then run the new-level suite.
- For I3+, O3+, SA4+, and U4+, require longitudinal evidence across multiple self/organizational changes rather than a single successful modification.
- For T5+, use sealed novelty environments, hidden mechanisms, or procedurally generated certification tasks to reduce memorization and benchmark leakage.
- For M1+ tests, include true episode boundaries or process/context restarts; do not allow the full prior transcript to remain in the active context when persistence is being measured.
- For M-Bench, measure relevance, provenance, temporal updates, contradiction handling, abstention, consolidation, repair/forgetting and downstream usefulness in addition to raw recall. For I-Bench, score the I-specific behavior separately and enforce the declared minimum M prerequisite.
- For I2+, require held-out behavioral transfer from consolidated experience; for I3+, preserve externally auditable change/version lineage so later self-improvement claims can be reconstructed from durable evidence.
- Audit benchmark items before model attribution: parameterized families must pass specification-key consistency tests, and concordant identical errors across materially different executors trigger item review.
- Report per-coordinate headroom and score distributions for graded item families; do not interpret a saturated coordinate as a model ceiling.
- For GUI/computer-use tasks, freeze screenshot resolution, observation channel, action interface and P_vis. Report GP diagnostics separately from end-to-end DI success.

19. What Should Remain Outside the Unified Core

Not every important property should be renamed as another intelligence. Several variables are essential for interpreting a system but are better treated as structural, environmental, or substrate coordinates.

Coordinate	Why it remains outside core intelligence
Circle / λ	Measures which substrate improvement loops are actually reached or closed; substrate access is not identical to cognition.
Write depth D	Measures which persistent cognitive states can be rewritten; permission to rewrite does not prove intelligent rewriting.
Authority A	Permission is not intelligence. A highly intelligent verifier may deliberately have almost no action authority.
Reach R	Access to files, tools, agents, robots, labs, or networks increases possible action but is not itself cognition.
Verification V	Determines credibility/independence of claims; it validates intelligence rather than constituting the tested intelligence.

Coordinate	Why it remains outside core intelligence
Separation S	Organizational validity condition for proposer/evaluator/promoter roles.
Resources	Compute, memory, money, time, energy and hardware affect achievable performance but should not inflate the intelligence definition.
Physical floors φ	Represents irreducible latency/physics constraints and degree of saturation rather than cognitive level.

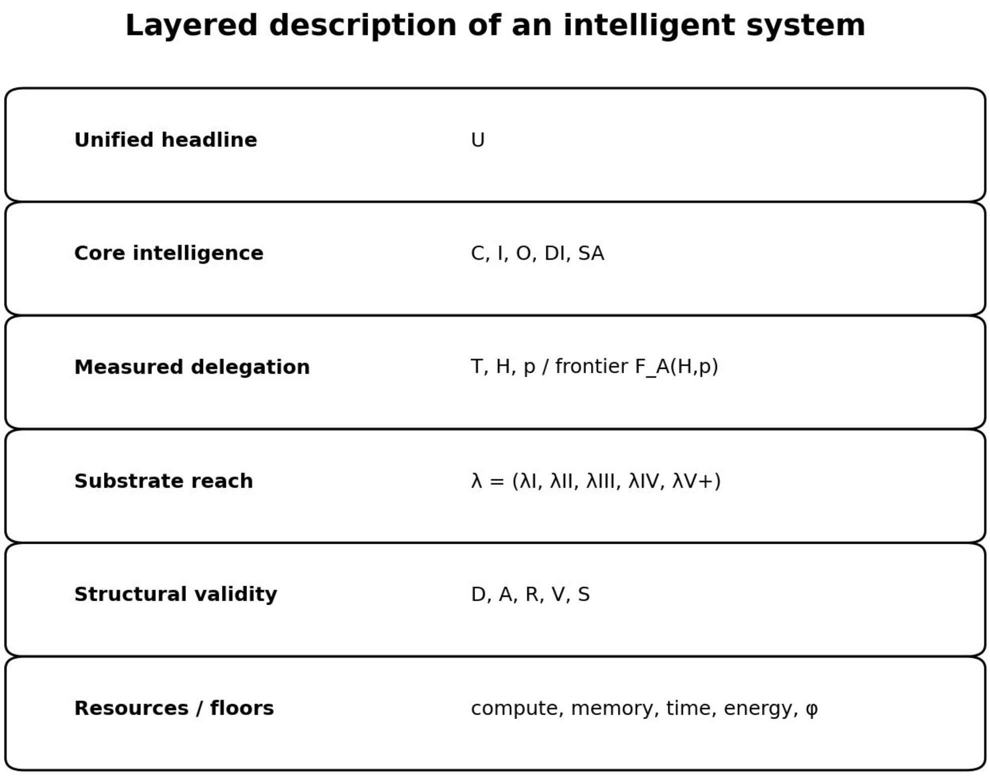

Figure 4. A complete system description layers the Unified headline over the core intelligence profile, measurable delegation frontier, substrate reach, structural validity conditions, and resource/physical constraints.

20. Worked Examples

20.1 Strong LLM, weak harness

$$[C5, I0, O0, T2, H2, SA1]$$

The model may solve difficult reasoning problems, but the instantiated agent has little persistent evolution and limited end-to-end delegation autonomy. High C does not automatically produce high U.

20.2 Moderate model, strong persistent harness

$$[C3, I2, O1, T4, H1, SA2]$$

This system can be highly useful operationally: persistence, planning, tools, retries, and verification allow difficult tasks to be delegated. Yet it is not self-improving in the I3 sense, and its self-model is not yet causal. The profile communicates these strengths and limits directly.

20.3 Advanced organization with a bottleneck

[C5, I4, O3, T4, H1, SA4] -> U3, bottleneck O3

Several dimensions exceed level three, but organizational intelligence has not reached O4. The unified level remains U3 until the organization demonstrates recursive improvement while retaining lower-level capabilities.

20.4 Same LLM Across a Harness Ladder

A base LLM is tested in HG0, HG1, HG2, and HG3. Its HLIS scores are 42, 55, 67, and 72; the corresponding certified levels are U0, U1, U2, and U3. The model therefore has HIL-Level 3 on the current standardized ladder, a ceiling of 72, and a large positive Harness Gain of 30 points. A second model might start at HLIS 60 in HG0 but rise only to 70 in HG3: it has stronger minimal-harness capability but lower harness gain. HIL-AUC and HIL-Ceiling distinguish these patterns rather than collapsing them into one prompt benchmark.

21. Implications for Intelligence Research

A practical consequence is that frontier-model evaluation should retain contemporary benchmark vectors rather than compress them prematurely. HIL adds a second axis of progress: harness scaling. This allows researchers to distinguish a new model that improves raw reasoning from a new harness that unlocks more of an existing model, and to identify models whose strengths are unusually composable into persistent agents or organizations.

The framework distinguishes model capability from agent capability and both from system capability. A benchmark that tests only answers under carefully prepared prompts primarily measures C. A benchmark that tests whether the same system can accept a difficult goal, maintain state, choose strategy, use tools, recover, verify, and finish with little human cognition measures the delegation surface. Longitudinal tests of learning and self-modification measure I, with long-term memory embedded as the temporal substrate: cross-session continuity at I1, consolidation and procedural transfer at I2, and improvement/discovery lineage at higher levels. Multi-agent role and reorganization tests measure O. Grounded introspective tests measure SA.

The pilot ladder adds a measurement principle: a score defined only on “how often was the produced artifact correct?” can miss differences that matter to delegation, such as whether incorrect work was confidently returned as finished or correctly held back. Outcome-complete evaluation should therefore preserve both task capability and delivery integrity rather than forcing them into one undifferentiated success bit.

This decomposition also reduces incentives to overclaim. A system can legitimately report high performance in one dimension without claiming a uniformly high intelligence level. Conversely, the cumulative U hierarchy creates a demanding standard for statements such as 'U4 intelligence': the system must retain lower capabilities and satisfy all required gates at the specified reliability.

The framework is intentionally extensible. Empirical work may revise thresholds or reveal that a currently grouped construct should split. However, new axes should be added only when a property is both operationally measurable and not adequately explained by the existing core or by structural/environmental coordinates.

The HIL framework also creates a new object of study: harnessability. Two LLMs with similar C-scores may differ substantially in their ability to exploit persistent memory, tools, multiple agents, verification loops, or self-improvement infrastructure. A standardized harness ladder can therefore reveal model properties that are invisible in single-turn benchmarks. It also lets researchers quantify whether a new harness architecture produces a general improvement across LLMs or only a model-specific effect.

Computer agents provide a useful example of why the framework separates capability, task difficulty and delegation. The ability to recognize and ground a screen is a cognitive sub-capability; the visual complexity of the screen is a property of the task; and the ability to finish the workflow with little human cognition is delegation. Collapsing these into one success rate hides whether progress came from a stronger model, a better visual harness, an easier interface, or better planning/recovery.

22. Limitations and Open Questions

- The proposed level thresholds are hypotheses and require empirical calibration across domains and models.
- Task difficulty T is multidimensional; a scalar T band is useful operationally but should retain an underlying difficulty vector for horizon, uncertainty, ambiguity, verification burden, novelty, and environmental change.
- Cognitive Intelligence C may itself require a broad psychometric/domain profile rather than one benchmark family.
- Self-awareness is operational and evidence-grounded; the framework does not address phenomenal consciousness.
- O-levels are only meaningful for organizations with real state/authority/evidence boundaries, not a single prompt that role-plays several agents.
- Open-ended levels require long evaluation horizons and are especially vulnerable to benchmark contamination, resource confounds, and vague success criteria.
- Higher U denotes broader validated capability, not moral value, desirability, safety, speed, or economic efficiency.
- HLIS and HIL weights are provisional. They require calibration against external outcomes, sensitivity analysis, and anti-gaming studies; leaderboard use should publish the full component profile.
- A standardized harness ladder must control tools, context budget, compute, retrieval corpus, authority, and verifier access. Otherwise HIL becomes a measure of infrastructure spending rather than LLM ability.
- Benchmark-version drift is a first-class threat. HIL reports must freeze benchmark versions, hidden splits, harness code, tool inventories, and resource envelopes; otherwise apparent model progress may come from changing evaluations.

- HIL can inherit weaknesses from the benchmarks it integrates. A benchmark vector with contamination, narrow domain coverage, or weak verifiers does not become valid merely by being placed inside HIL.
- Harness Scaling Curves can be non-monotonic. A more complex harness may hurt a model through context overload, tool confusion, or protocol mismatch; HIL should report regressions rather than force monotonic scores.
- A composite HIL-Score should never be used without the benchmark vector and component profile. Different users may rationally prefer different points on the capability/autonomy/cost frontier.
- Long-term-memory evaluation is vulnerable to leakage and trivialization. If a benchmark leaves the complete history in the active context, it may measure long-context reading rather than persistent Memory Capability; if it measures recall only, it can reward stale or irrelevant storage. M-Bench therefore needs true temporal discontinuities, update/contradiction cases, provenance checks, consolidation/repair tests, and downstream-usefulness measures. The proposed mu_n prerequisites for I are working definitions and require empirical calibration.
- The pilot measured only Delegation Intelligence at H1 over a small HG0-HG3 ladder. It does not establish full HLIS or Unified Intelligence, and the upper HG4-HGΩ ladder remains uninstantiated in the reported pilot.
- The initial frontier task suite showed saturation for stronger executors; identical aggregate curves can therefore indicate insufficient benchmark headroom rather than model equivalence.
- Loss-weighted delegation utility depends on predeclared task-class cost ratios. The delivered-correct frontier, false-completion rate, false-rejection rate, human load, and raw outcome table should always remain available so application-specific utility assumptions can be changed without rerunning the model.
- Verification Responsiveness is sensitive to feedback quality and criterion granularity; comparisons require standardized, accurate failure messages.
- GUI-perception measurements are highly sensitive to observation channel, resolution, scaling, accessibility metadata and action representation. Cross-model comparisons must freeze these or explicitly treat them as harness differences.
- GP is intentionally a diagnostic C subscale rather than a sixth core coordinate. If future evidence shows a perceptual construct that is operationally independent of C and predicts distinct HIL behavior, the framework can revisit that choice.
- Item-validity audits reduce but cannot eliminate benchmark error. Cross-tier concordance is a trigger for review, not proof that the key is wrong; independent expert analysis and executable consistency tests remain necessary.

23. Conclusion

A unified intelligence level is useful only if it summarizes rather than erases structure. This paper therefore retains five conceptual intelligence families - Cognitive, Individual, Organizational, Delegation, and Self-Awareness Intelligence - while measuring Delegation through Task Difficulty and Human Cognitive Intervention at verified reliability. Version 1.8 reports Memory Capability M independently as a supporting coordinate rather than identifying M one-to-one with I. GUI perception remains inside C; visual-interface complexity remains a task property.

Version 1.7 also strengthens the measuring instrument. A HIL claim is only as credible as the items beneath it: parameterized specifications must agree with their keys, concordant cross-tier errors must be audited, graded items should preserve partial information and failure modes, and saturated coordinates must report their lack of headroom. For computer agents, the same discipline requires separating screen perception and grounding from closed-loop task execution. The result is a framework that can say not only how well a harnessed model performs, but where its realized intelligence is limited and whether the benchmark is capable of seeing that limit.

For harness-based AI, the framework provides both a development roadmap and a measurement program. Existing benchmark families remain visible as a capability vector; HIL adds the Harness Scaling Curve by freezing the LLM and progressively strengthening the harness. Memory is now independently measured, allowing the curve to reveal whether a model can exploit a strong memory subsystem without conflating that subsystem with learning or self-improvement. HLIS remains based on C/I/O/DI/SA so the same memory evidence is not counted twice; M-level and optional A_M are reported beside it. Outcome integrity, item validity, and benchmark headroom remain mandatory companion measurements.

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Appendix A. Compact Reporting Template

Core measured profile: [C, I, O, T, H, SA] @ reliability p

Supporting memory profile: M_m (and optional A_M); conceptual profile remains [C, I, O, DI, SA], where DI = F_A(H,p)

Unified headline: U_n [C_x, I_y, O_z, T_k/H_h, SA_s] | M_m

Extended system profile: [U; C,I,O,DI,SA | M; lambda; D,A,R,V,S; resources, phi]

Field	Example
Unified headline	U3
Core profile	C5, I4, O3, T4/H1, SA4
Reliability	p = 0.80
Delegation frontier	F_A(H0,.80)=T3; F_A(H1,.80)=T4
Bottleneck	O3
Circle reach	λ = (.42,.08,.03,0,0)
Structural profile	D3, A2, V4, separated proposer/evaluator/promoter
Resource profile	Declared compute, time, memory, energy/cost budget
Supporting memory capability	M4 (example); episodic/provenance 0.91, temporal update 0.84, consolidation 0.76; reported outside HLIS
Outcome integrity	False-completion rate=0.00; false-rejection rate=0.01; VR=0.72 (example)
GUI perception (when applicable)	GP level/profile; observation channel; grounding/state metrics.
Visual task complexity	P_vis with task-difficulty vector; resolution/window/dynamic-state notes.
Item-validity status	Headroom; concordance audit; spec-key consistency; graded-score distribution.

Appendix B. Suggested Unified Promotion Tests

Transition	Minimum new evidence in addition to full lower-level retention
U0 -> U1	Persistent cross-session state and episodic/task-memory recovery with provenance, temporal ordering, stale-state handling, and correct goal-level completion after restart/context discontinuity.
U1 -> U2	Verified experience is consolidated into durable semantic/procedural knowledge that improves held-out related tasks; knowledge updates/contradictions are handled; multi-step delegation under <=H2.
U2 -> U3	Governed self/organizational method improvement plus durable failure/hypothesis/change/benchmark/version lineage, T3 strategy construction under <=H1, and causal self-diagnosis.
U3 -> U4	Validated improvement of the improvement process using longitudinal meta-improvement history, plus long-horizon T4 work and accurate modeling of self-change.
U4 -> U5	Independent discovery of initially unknown solution method/knowledge with persistent hypothesis-experiment-evidence lineage and self-directed experiments about world and self-unknowns.

Transition	Minimum new evidence in addition to full lower-level retention
U5 -> U6	Mission -> projects -> tasks -> execution loop over changing conditions with persistent role/mission/decision history and grounded mission self-model.
U6 -> UΩ	Repeated bounded-charter mission generation whose validated discoveries create new tools/agents/organizations, with governed evolution of memory/cognitive instruments and preserved external lineage.

Framework status. This document is a theoretical proposal developed from the preceding Unified Intelligence, Delegation Intelligence, harness-indexed intelligence, and operational self-awareness design discussion. Level names and numerical thresholds are proposed working definitions and should be revised when benchmark evidence warrants.

Appendix C. Harness-Intelligence Reporting Template

Base LLM: model id/version/hash

Standardized harness ladder tested: HG0, HG1, ..., HGk

Per-harness: U* | HLIS | [C,I,O,T/H,SA] | M-level/A_M | delivered-correct reliability | bottleneck | false-completion/false-rejection rates | VR | resource budget | benchmark/ladder id

LLM summary: benchmark vector B(m) | Harness Scaling Curve HSC | HIL-Level | HLIS(HG0) | HIL-Ceiling | Harness Gain | VR | HIL-AUC/composite if reported with cautions | HDS if harness-design tasks were run

Application panel: selected coordinates such as I, O, SA, Circle lambda, verification strength, cost/latency - reported separately from the common core.

Appendix D. Version History

Version	Principal change
1.0	Five families, six measured coordinates, cumulative U0-UΩ hierarchy, HG0-HGΩ concept.
1.1	HLIS for a model-harness pair; HIL characterization across a standardized harness ladder.
1.2	Contemporary benchmark mapping and Harness Scaling Curve.
1.3	Formal HLIS construction, normalization, cumulative achievement, confidence/resource reporting.
1.4A	Long-term memory embedded inside Individual Intelligence; longitudinal I-Bench and memory diagnostics.
1.4B	Parallel measured-ladder draft: HG0-HG3 pilot, acceptance-

Version	Principal change
	invariance finding, false-completion-aware scoring, verification responsiveness, paired-rung protocol.
1.5	Merged revision: memory-integrated I plus outcome-complete delegation measurement, measured-ladder lessons, benchmark binding status, and curve-first HIL reporting.
1.6	Benchmark-item validity: cross-tier concordance audit, specification-key consistency tests, graded item scoring with separated failure modes, harder DL4 item families, and per-coordinate headroom.
1.7	Merged from Claude Code v1.6 and HIL v1.5.1; adds GUI Perception GP inside C, visual-interface complexity P_vis inside task difficulty, computer-use diagnostics inside DI, and GUI-aware benchmark/reporting rules.
1.8	Memory Capability M becomes an independently measured supporting coordinate with M0-MΩ and M-Bench; Individual Intelligence uses one-way prerequisite gates I_n => M>=mu_n; memory may exceed I; A_M is reported separately and excluded from default HLIS to prevent double counting.